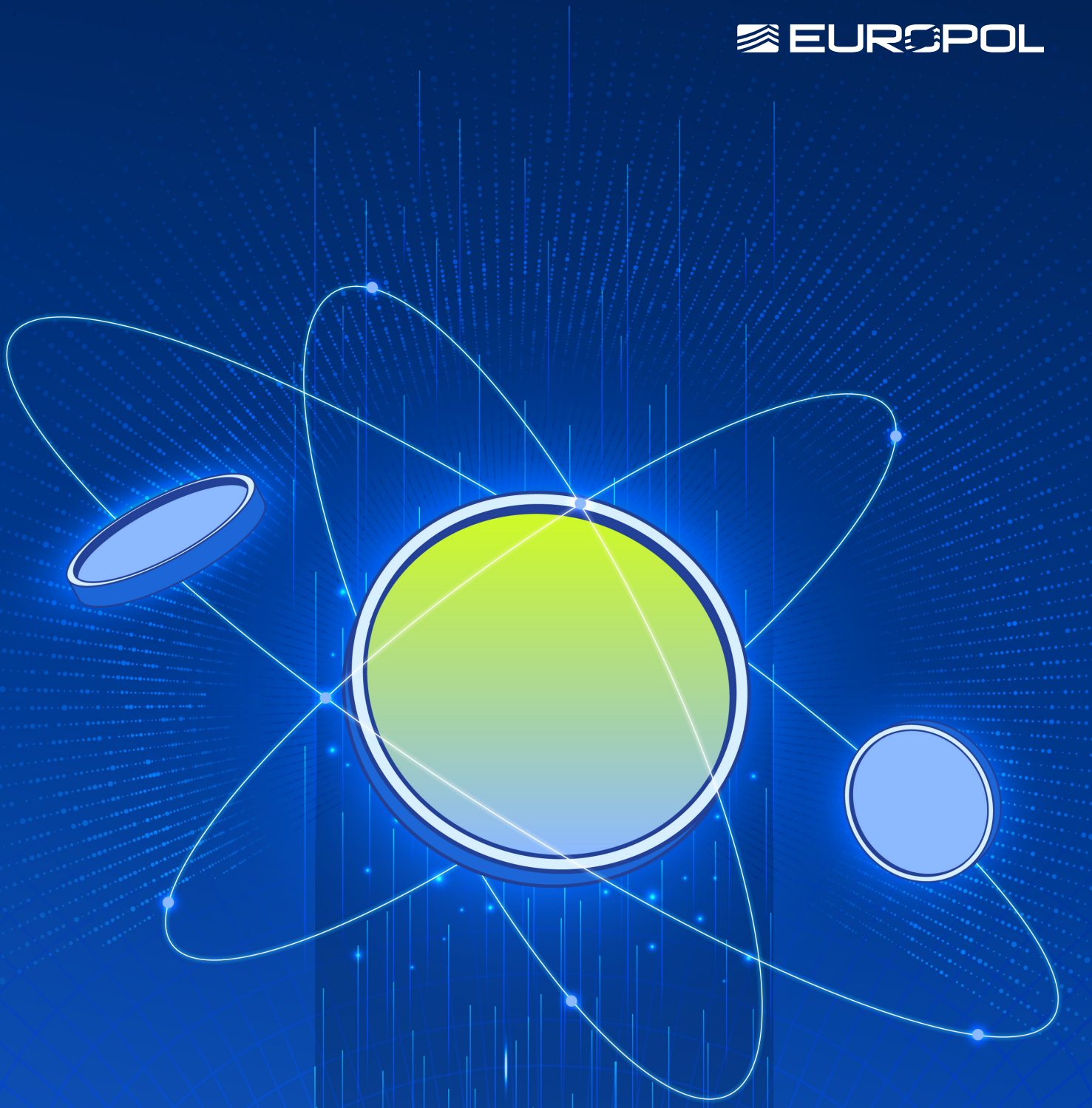
EUROPOL



QUANTUM COMPUTING AND CRYPTOCURRENCIES

Bridging technical expertise and decision-making

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Executive summary

This report by Europol's European Cybercrime Centre (EC3) evaluates the potential implications of quantum computing on the cryptocurrency landscape. The report delivers an in-depth technical analysis that pinpoints genuine risks, recommends mitigation strategies, and addresses misconceptions that could influence the future of cryptocurrencies.

The report builds on Europol's existing work on quantum technologies, including *The Second Quantum Revolution: the impact of quantum computing and quantum technologies on law enforcement*[1], *the Quantum Safe Financial Forum – A call to action*[2], and the *joint report on Prioritising Post-Quantum Cryptography Migration in Financial Services*[3]. These series of publications focus on raising awareness, supporting preparedness, and promoting the transition towards post-quantum cryptography.

Key findings

01. Quantum computing poses **a credible threat to current public-key cryptography**, potentially impacting the security of digital assets.

02. The blockchain's core integrity, secured by cryptographic hash functions, remains largely resistant to quantum attacks. **The primary risk lies in the vulnerability of cryptocurrency wallets** that rely on public-key cryptography.

03. **Proactive adaptation**, rather than systemic collapse, is the most likely outcome. Cryptocurrencies can evolve through incremental upgrades, adoption of post-quantum cryptography (PQC) and exploration of quantum-enhanced security measures.

1 https://www.europol.europa.eu/publications-events/publications/second-quantum-revolution-impact-of-quantum-computing-and-quantum-technologies-law-enforcement

2 https://www.europol.europa.eu/publications-events/publications/quantum-safe-financial-forum-call-to-action

3 https://www.europol.europa.eu/media-press/newsroom/news/joint-report-outlines-practical-approach-to-prioritising-post-quantum-cryptography-migration-in-financial-services

Primary risk

Cryptocurrency wallets, which utilise public-key cryptography to authorise transactions and control funds, are the primary point of exposure to quantum threats. Once sufficiently powerful quantum computers are developed, these keys could be compromised, leading to unauthorised access and potential loss of funds. However, solutions already exist.

Mitigation strategies

- **Incremental upgrades**: implement account abstraction and multi-signature schemes to reduce reliance on single public keys and enhance wallet security.
- **Post-quantum cryptography (PQC) adoption**: transition to NIST-standardised PQC algorithms (e.g. CRYSTALS-Kyber, CRYSTALS-Dilithium) to replace vulnerable cryptographic schemes.
- **Quantum-enhanced security**: explore the potential of quantum key distribution (QKD) and other quantum technologies to further strengthen blockchain security, though scalability remains a challenge.

Recommendations

- **Decision makers and regulators**: establish clear guidelines for PQC adoption and incentivise industry collaboration on quantum-resistant solutions.
- **Blockchain projects**: prioritise the integration of PQC algorithms into core protocols and provide resources for wallet providers to implement these changes.
- **Wallet providers**: begin testing and deploying PQC-enabled wallets, and educate users about the risks and benefits of quantum-resistant cryptography.
- **Users**: stay informed about the evolving quantum threat and proactively migrate funds to quantum-resistant wallets as they become available.

Conclusion

Cryptocurrencies will not collapse due to quantum computing, but their long-term security requires proactive defence. By embracing innovation, fostering collaboration and prioritising a phased transition to quantum-resistant cryptography, the cryptocurrency ecosystem can build a safer, more resilient and trustworthy future. The time to act is now, to lay the foundation for a quantum-resistant digital economy.

I. Introduction

Cryptocurrencies have rapidly evolved from a niche technological experiment to a globally recognised asset class and a platform for decentralised applications. This ascent has been accompanied by increasing scrutiny, not only from financial regulators and traditional institutions, but also from a growing chorus of commentators highlighting potential vulnerabilities. Among these concerns, the threat posed by the development of quantum computing has gained significant traction, frequently appearing in headlines and fuelling narratives of impending obsolescence for digital currencies. While quantum technologies promise transformative advancements across diverse sectors – from accelerating drug discovery and materials science to revolutionising financial modelling – they also introduce a fundamental challenge to the cryptographic foundations upon which modern digital security, and cryptocurrencies specifically, are built.

The core of this concern lies in the potential for quantum computers to break widely used public-key encryption algorithms. These algorithms, such as Rivest–Shamir–Adleman (RSA) and elliptic curve cryptography (ECC), underpin the security of countless digital transactions, communications and data storage systems. However, the advent of sufficiently powerful quantum computers, leveraging algorithms like Shor's, could render these algorithms ineffective, potentially exposing sensitive data and undermining trust in digital infrastructure. This has led to widespread speculation about the future of cryptocurrencies, with some predicting their complete collapse in the face of quantum attacks.

However, such a conclusion is overly simplistic. Cryptocurrencies are not solely reliant on public-key cryptography. They also employ symmetric cryptography and, crucially, cryptographic hash functions, which exhibit a different security profile. While public-key schemes are vulnerable to quantum algorithms, hash functions, used extensively in blockchain integrity mechanisms such as proof-of-work and proof-of-stake, are considered more resilient, although not entirely immune, to quantum attacks. Therefore, the threat is not necessarily the complete eradication of cryptocurrencies, but rather a targeted vulnerability in the ownership and control of digital assets – specifically, the security of cryptocurrency wallets.

This report undertakes a comprehensive analysis of the impact of quantum computing on cryptocurrencies. We will clearly delineate the extent of the quantum threat to cryptography, detailing the algorithms at risk and the timeline for potential breakthroughs. We will then examine how cryptocurrencies utilise cryptography, highlighting the interplay between vulnerable and resilient cryptographic primitives. Crucially, we will propose a range of mitigation strategies, from incremental upgrades to the adoption of post-quantum cryptographic standards, and explore the potential for leveraging quantum technologies to enhance blockchain security. Finally, we will present a reasoned conclusion based on our analysis and offer actionable recommendations to policymakers, industry stakeholders and users, with a view to facilitating a proactive and secure transition to a quantum-resistant future for cryptocurrencies.

This report aims to move beyond sensationalism and provide a nuanced, evidence-based assessment of the challenges and opportunities presented by the convergence of quantum computing and blockchain technologies.

II. Quantum computing

Quantum computing was born in the early 1980s4. It is expected to have as much impact, if not more, as previous technological developments based on quantum mechanics (such as the transistor in 1947, the atomic clock in 1949 or the laser in the mid-1950s). However, these new developments could pose a threat, if they are not accompanied by the corresponding mitigation.

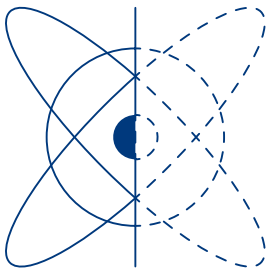
To understand and demystify the threat that quantum computing poses to cryptocurrencies, it is crucial to understand the basic principles of quantum computing, its current state of development and the impact it will have on the technologies on which cryptocurrencies are based.

a. Principles behind quantum computing

Quantum computing[5] is an emerging class of computing technology that operates on fundamentally different principles from classical computers. While classical computers process information using bits that are either 0 or 1, quantum computers use quantum bits, also called *qubits*, which can represent both states (0 and 1) simultaneously according to the quantum mechanical phenomena known as **quantum superposition**.

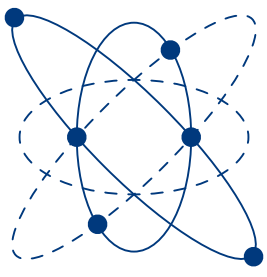
Similarly to the Schrödinger's cat thought experiment, those quantum particles convey both states simultaneously. However, measuring (observing) a qubit will make this superposition collapse to a single state (0 or 1), somehow converting the qubit to a regular bit.

p. I



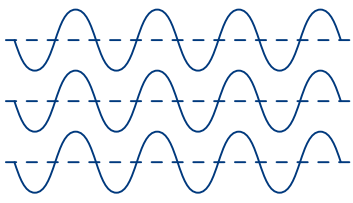
quantum superposition

p. II



quantum entanglement

p. III



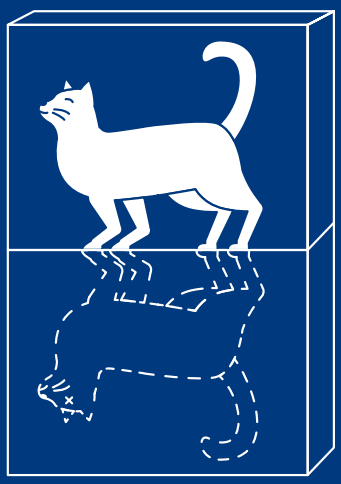
quantum interference

4 Benioff, Paul, 'Quantum mechanical Hamiltonian models of Turing machines', Journal of Statistical Physics, 1982

5 Ghose, Shohini (IBM), A beginner's guide to quantum computing – video recording of TEDWomen conference

Schrödinger's cat is a thought experiment devised in 1935. Imagine a cat sealed in a box with a radioactive atom, a Geiger counter, and a vial of poison. If the atom decays, the counter triggers and the poison is released, killing the cat. If it doesn't decay, the cat lives. According to quantum mechanics, before you open the box, the atom is in a superposition of decayed and not decayed — which implies the cat is both alive and dead at the same time.

Schrödinger invented this scenario to mock the idea, not to promote it. He wanted to show how absurd it is to apply quantum superposition to everyday objects like cats. Ironically, the thought experiment became one of the most famous illustrations of quantum theory, even though it was originally meant as a critique of its apparent irrelevance to the macroscopic world.



Another crucial quantum property in quantum computing is the property of **quantum entanglement**. Two or more entangled particles become linked in such a deep way that they act like a single system, even when they are far apart[6]. Entanglement lets qubits work together instead of separately, and measuring one particle instantly tells you something about the other one.

Quantum computers exploit superposition and entanglement to represent and manipulate a large space of possible states. However, these properties alone are not sufficient to achieve a quantum speedup. The crucial mechanism is **quantum interference**, through which a quantum algorithm is designed to amplify the probability of correct outcomes while suppressing incorrect ones during the computation.

When the system is finally measured, the quantum state collapses to a single outcome, ideally corresponding to the expected result. This process is inherently probabilistic; quantum algorithms must increase the likelihood of obtaining the correct answer, but do not guarantee it in a single run. Quantum superpositions cannot be directly observed or 'read', which fundamentally limits how quantum computations can be used and interpreted.

In practical terms, this allows quantum computers to perform **certain** categories of calculations much more efficiently, sometimes with an exponential gain in complexity, than classical computers. It remains important to understand that such a gain is not universal, however, and only applies to specific algorithms. Nevertheless, unlike incremental improvements in classical computing, quantum computing represents a discontinuous technological shift. It introduces new computational capabilities rather than simply faster versions of existing ones.

6 Horodecki, R., Horodecki, P., Horodecki, M., & Horodecki, K., 'Quantum entanglement', Reviews of Modern Physics, 81(2), 2009, pp. 865–942, https://doi.org/10.1103/revmodphys.81.865

b. Status of quantum computers

The European Union and countries worldwide are investing heavily in quantum computing, supporting a wide range of technologies and infrastructures as listed in *Table 1* below. Through initiatives like the EU Quantum Technologies Flagship[7] and EuroHPC JU[8], the European Union has developed working prototypes, software tools and spin-offs, while also deploying early quantum systems across member states. These efforts aim to foster a sovereign, competitive quantum industry and expand the market for both hardware and software solutions.

Technology platform	Superconducting	Ion Traps	Cold atoms	Photonics	Spin Qubits
EU machines	17	6	8	5	3
UK machines	4	6	0	5	2
USA machines	26	7	4	2	0
Canada machines	13	0	0	1	0
China machines	1	0	0	1	3

Table 1: Quantum computing and simulation suppliers landscape (source: Quantum Europe Strategy: Quantum Europe in a Changing World[9])

Quantum computers can be built using different technological approaches. There are many different ways to interact with, generate and manipulate subatomic particles, such as, using superconductive materials, laser beams and photonic circuits[10]. Maintaining these systems can be technically challenging and costly, though the specific requirements vary by platform. Superconducting qubits, for example, requires extremely low temperatures (close to absolute zero) to maintain coherence. Trapped-ion and neutral-atom qubits typically operate at room temperature or in slightly cool conditions, but require precise laser control and ultra-high vacuum systems. Photonic qubits can also function at ambient temperatures but demand sophisticated optical setups. In all cases, the delicate nature of qubits makes quantum computing hardware significantly more difficult to build and maintain than classical systems, which constitutes a serious obstacle to quantum computers' calculations.

Notably, noise, both environmental and internal, represents a big challenge[11] because, regardless of the processors' power, it significantly multiplies computational error, therefore directly inhibiting the computer's operations. On one hand, because many computers operate at above-zero temperatures (to preserve quantum states), thermal radiation and other quantum phenomena affect qubits' calculations and lead to their degradation. On the other hand, error multiplication also derives from the nexus between physical

7 Quantum Flagship

8 The European High Performance Computing Joint Undertaking (EuroHPC JU)

9 COM(2025) 363 final - Quantum Europe Strategy: Quantum Europe in a Changing World

10 Qubits, Institute for Quantum Computing, University of Waterloo

11 Quantum computing is taking on its biggest challenge: noise, MIT Technology Review

and logical qubits. Physical qubits are the raw, error-prone units of a quantum computer (such as ions or superconducting circuits), while logical qubits are error-corrected, reliable units formed by combining multiple physical qubits. In other words, logical qubits are used for running algorithms, while extra physical qubits are dedicated to error correction and redundancy. As we add more logical qubits, the number of possible errors grows faster than the number of qubits themselves. Each logical qubit requires many physical qubits and extra operations for error correction, meaning that the total number of errors can grow in a non-linear way[12].

Quantum error correction is a critical milestone for any company or start-up developing a quantum computer. In 2021, Google announced its goal of building a useful, error-corrected quantum computer within the decade[13], while IBM's Quantum Roadmap sets a target of a fault-tolerant quantum computer by 2029. However, the timeline for when quantum computers will be powerful enough to solve currently intractable problems remains uncertain. Although recent research claims to have achieved quantum supremacy[14], the practical relevance of the problems solved is still debated. The 2025 Quantum Threat Timeline Report[15] surveyed 32 global experts from academia and industry. *Figure 1* summarises their expectations as regards when a quantum computer could break RSA-2048, a widely used cryptosystem, within a reasonable timeframe (24 hours).

Thus, quantum computers are likely to become a reality in the medium to long term. It is therefore critical to assess the risks they pose to society and, in the context of this document, to the cryptocurrency ecosystem in particular.

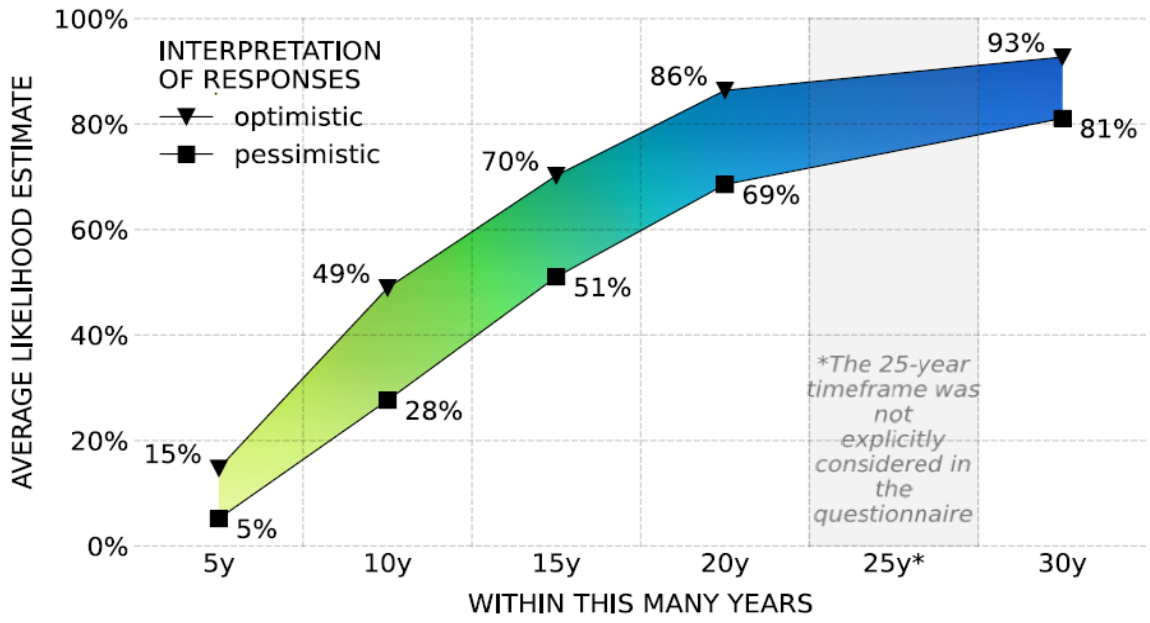


Figure 1: 2025 Opinion-based estimates of the likelihood of a digital quantum computer able to break RSA-2048 in 24 hours, as function of time. (source: Quantum Threat Timeline Report 2025 - EvolutionQ, GRI (4))

12 Google Quantum AI, 'Suppressing quantum errors by scaling a surface code logical qubit', Nature 614, 2023, pp. 676–681, https://doi.org/10.1038/s41586-022-05434-1

13 Erik Lucero, 'Unveiling our new Quantum AI campus', Google AI Blog, 18 May 2021, https://blog.google/technology/ai/unveiling-our-new-quantum-ai-campus/

14 Quantum supremacy refers to the point at which a quantum computer can solve a problem that is practically impossible for classical computers to solve, even with extensive resources. It demonstrates the quantum computer's ability to outperform classical systems in specific tasks.

15 Quantum Threat Timeline Report 2025, Global Risk Institute

c. Cryptographically-relevant quantum capabilities

Quantum technologies already have a significant impact on everyday life[16]. Well-known applications of quantum mechanics include semiconductor electronics, laser technology and magnetic resonance imaging (MRI). Beyond these established uses, ongoing progress in areas such as quantum sensing, quantum optimisation and quantum communication[17] is expected to enable a wide range of new technological capabilities.

This report focuses on cryptocurrencies and therefore restricts its scope to quantum technologies that are relevant to this context. In particular, it examines the potential impact of quantum computing on cryptographic algorithms that underpin the security of modern cryptocurrencies. There exist today two important algorithms with a potential concrete impact on current standards cryptography: Shor's and Grover's algorithms.

SHOR'S ALGORITHM

Shor's algorithm[18] is widely regarded as the most significant quantum algorithm in the context of cryptography. It enables the efficient factorisation of large integers and the computation of discrete logarithms. Those problems are believed to be computationally unpractical for classical computers. Most of the classic public-key cryptographic systems, including RSA, Diffie-Hellman and elliptic curve cryptography, rely directly on the difficulty of these problems to ensure our modern digital-security. These systems are widely used today to secure web communications (TLS), software updates, digital identities and blockchain-based signature schemes.

The existence of Shor's algorithm implies that a sufficiently powerful and fault-tolerant quantum computer would be able to break these cryptographic schemes in polynomial time. In practical terms, this would allow an attacker to derive private keys from public information, enabling the forging of digital signatures or the decryption of previously secure communications. Although such quantum computers do not yet exist, this potential capability represents a fundamental threat to current public-key infrastructures and is the main driver behind the development, standardisation and transition to post-quantum cryptographic standards. Indeed, current data may already be at risk according to Mosca's inequality theorem[19], which warns that data encrypted today could be decrypted in the future, threatening potential long-term confidentiality requirements.

16 Scholten, T. L., et al., 'Assessing the Benefits and Risks of Quantum Computers', arXiv preprint, 2024, https://arxiv.org/abs/2401.16317

17 Europol, 'The Second Quantum Revolution: the impact of quantum computing and quantum technologies on law enforcement', Europol Innovation Lab Observatory Report, Publications Office of the European Union, 2023, https://www.europol.europa.eu/publications-events/publications/second-quantum-revolution-impact-of-quantum-computing-and-quantum-technologies-law-enforcement

18 Shor, P. W., 'Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer', SIAM Journal on Computing, Vol. 26(5), 1997, pp. 1484–1509. https://doi.org/10.1137/s0097539795293172

19 Michele Mosca, 'Cybersecurity in a quantum world: will we be ready?', https://www.youtube.com/watch?v=VXh6xLrP7ys

GROVER'S ALGORITHM

In contrast, Grover's algorithm[20] affects cryptography in a more limited but still relevant way. It provides a quadratic speedup for unstructured search problems, including brute-force attacks against symmetric encryption schemes and cryptographic hash functions.

These primitives are commonly used to protect stored data, secure communications channels and ensure data integrity, for example through encryption standards such as AES or hash functions used in password storage and blockchain mining.

A **cryptographic hash** function is a mathematical function that takes an input of arbitrary size (such as a file, a message, or a transaction) and produces a fixed-size output called a hash or digest. This output appears random, but it is completely determined by the input: the same input will always produce the same hash.



Hash functions are designed to be **one-way**: given a hash, it is computationally infeasible to recover the original input. They are also **collision-resistant**, meaning it is extremely difficult to find two different inputs that generate the same hash. These properties make hash functions useful for verifying data integrity and securing digital systems.

In cryptocurrencies, hash functions are fundamental. They are used to identify transactions and blocks, to link blocks together in a blockchain and to support consensus mechanisms such as proof-of-work. The security of these systems relies on the assumption that hashes cannot be predicted, reversed or efficiently manipulated.

Rather than completely breaking these systems, Grover's algorithm effectively reduces their security level. In practice, its impact can be mitigated by increasing key and hash sizes, for instance by moving from 128-bit to 256-bit keys. As a result, symmetric cryptography is generally considered more resilient to quantum computing than public-key cryptography, provided that the appropriate parameter choices are adopted.

Beyond these specific algorithms, quantum computing may also enhance general cryptanalytic capabilities. Improved search and optimisation techniques could facilitate the analysis of cryptographic constructions, including the new post-quantum cryptographic standards, or lower the cost of certain attacks. In the context of cryptocurrencies, where cryptography underpins ownership, transaction validation and consensus mechanisms, these developments raise important questions about long-term security and trust, and naturally motivate a closer examination of how quantum computing may affect cryptocurrency systems.

20 Grover, L. K., 'A fast quantum mechanical algorithm for database search', STOC '96: Proceedings of the twenty-eighth annual ACM symposium on Theory of Computing, 1996, https://doi.org/10.1145/237814.237866

III. Cryptocurrencies and their security foundations

Cryptocurrencies are often perceived primarily as financial instruments or investment assets. While this is partially true, at their core they are cryptographic systems designed to provide **secure, decentralised** and **verifiable transactions without relying on traditional central authorities.** The security and reliability of these systems depend on mathematical and cryptographic principles, rather than institutional oversight. As a result, any shift in the underlying assumptions of cryptography, as discussed in the previous section, has the potential to impact cryptocurrencies in fundamental ways[21]. Before proceeding to a systematic review of the associated risks and countermeasures, it is useful to provide a brief reminder of the core principles underlying cryptocurrency systems.

Unlike traditional financial systems, where trust is placed in institutions, cryptocurrencies derive trust from **mathematical guarantees.** Transactions are validated through consensus mechanisms and recorded in a public ledger, known as a blockchain. The blockchain ensures that transactions cannot be altered once confirmed, providing transparency and security in a decentralised manner.

a. Proof-of-work, proof-of-stake and blockchain integrity

Cryptocurrencies rely on different consensus mechanisms to validate transactions and secure the blockchain, the most common being **proof-of-work (PoW)** and **proof-of-stake (PoS).**

In PoW systems, such as Bitcoin, miners compete to solve computational puzzles based on cryptographic hash functions. These puzzles require finding an input that produces a hash output meeting a certain condition, such as a number of leading zeros. Hash functions are easy to compute in one direction but extremely difficult to invert, ensuring that altering a block in the ledger would require infeasible amounts of computation. Importantly, the security of PoW depends on this computational difficulty rather than on secrecy; it guarantees that tampering with past blocks is prohibitively expensive.

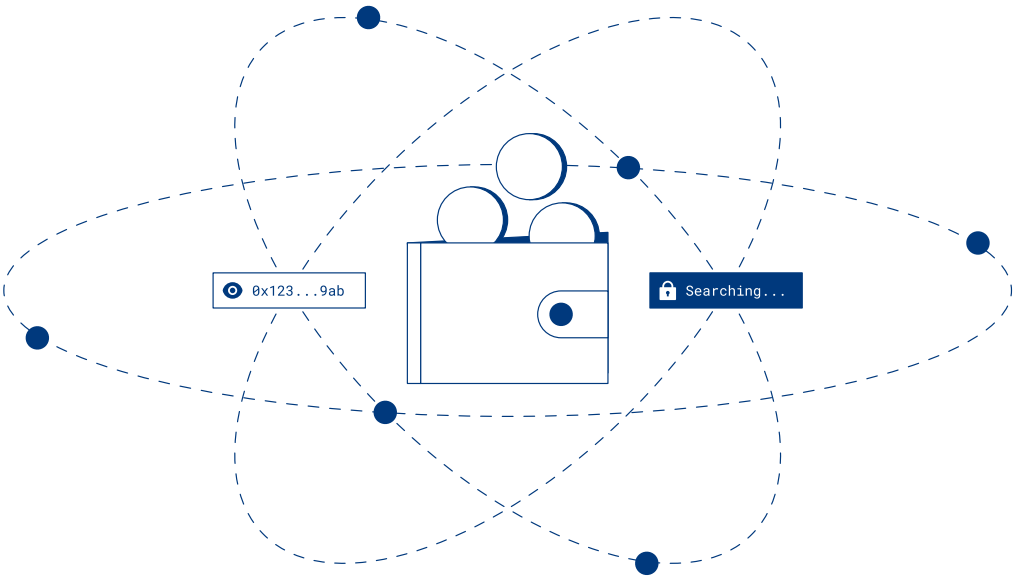
PoS systems, like Ethereum post-merge, use a different approach. Validators are chosen to propose and confirm blocks based on the amount of cryptocurrency they 'stake' as collateral. While PoS does not rely on computational puzzles for security, hash functions still play a critical role in linking blocks and maintaining the integrity of the blockchain. Every block contains a hash of the previous block, forming a chain that is computationally secure against alteration. Because these hash functions are

21 Kappert, N., Karger, E., & Marko Kureljusic, 'Quantum Computing – The Impending End for the Blockchain?', Pacific Asia Conference on Information Systems (PACIS) 2021, Dubai, UAE, 114

resistant to inversion, the blockchain remains secure even under PoS, despite the absence of mining-based work.

In both PoW and PoS systems, hash functions are **largely quantum-safe**[22]. Quantum computers could accelerate certain search problems through algorithms such as Grover's algorithm, providing roughly a square-root speedup in finding hash preimages or collisions. For example, compromising a 256-bit hash would require on the order of 2^{128} operations for a quantum computer, still astronomically high with foreseeable technology. This contrasts sharply with public-key cryptography, which will be fully broken by quantum algorithms such as Shor's, and therefore requires separate mitigation strategies.

In short, the use of hash functions in both PoW and PoS ensures that the **integrity of the blockchain itself remains secure against quantum attacks**, making these mechanisms a robust foundation for decentralised networks, at least in the medium term. The primary quantum risk, as discussed in the next subsection, lies elsewhere: in the cryptography securing ownership and wallets.



b. Wallet security and public-key cryptography

While the blockchain ledger itself is secured by hash functions and consensus mechanisms, the most sensitive part of a cryptocurrency system is the ownership of digital assets. This ownership is established and maintained through public-key cryptography, which functions similarly to a decentralised form of public-key infrastructure (PKI).

Every cryptocurrency wallet relies on a pair of cryptographic keys: a private key and a public key. The private key is kept secret and allows the owner to authorise transactions, proving control over the associated funds. The public key, in contrast, can be shared openly and is used by the network to verify that any transaction claiming to spend funds from that wallet is valid. The cryptographic relationship between the private and public key ensures that the private key cannot be deduced through classical computation, thereby securing the wallet.

It is important to note that PKI in cryptocurrencies is not used to validate the blockchain itself. When

22 Bernstein, D., Lange, T., 'Post-quantum cryptography', Nature, 549, 2017, pp. 188–194, https://doi.org/10.1038/nature23461

a miner discovers a new block in a proof-of-work system, or when a validator proposes a block in a proof-of-stake system, the network verifies the block's validity through hash functions and consensus rules, not through general-purpose digital signatures such as PKI. In PoW, the network only checks that the submitted proof-of-work meets the difficulty criteria, while in PoS, the protocol ensures the validator is eligible based on staked assets. Any signature included with a block is typically protocol-specific and serves to authenticate the proposer, not to secure funds.

The distinction is critical from a quantum perspective. While hash functions securing block integrity are currently quantum-resistant, the cryptographic keys used in wallets are vulnerable to sufficiently powerful quantum computers. Algorithms such as Shor's algorithm will derive a private key from a public key, allowing an attacker to spend funds without authorisation. This is exactly why **wallet security**, rather than blockchain integrity, is considered the **primary point of exposure to quantum threats**.

From a policy standpoint, this creates several important considerations. Unlike traditional financial systems, there is no central authority that can revoke compromised keys or enforce upgrades. Decentralised governance and long-lived addresses mean that vulnerabilities could persist for years if the migration to quantum-resistant cryptography is not carefully planned and coordinated. Moreover, the public visibility of blockchain keys enables adversaries to proactively target wallets, creating a latent risk of future attacks once quantum computers become capable, unless decisive action is taken to transition away from the status quo.

In summary, the **primary quantum risk** lies in **public-key cryptography** used by wallets, not the blockchain itself, which remains secure thanks to hash functions. The focus must therefore shift to practical mitigation: improving key management, educating users, upgrading protocols, and coordinating large-scale migration to quantum-resistant standards.

c. Decentralised governance and migration challenges

Cryptocurrencies replicate many functions of traditional public key infrastructure (PKI) in a decentralised way, where identity verification and transaction authenticity are mathematically guaranteed rather than institutionally enforced. While this design enhances independence, it complicates critical upgrades, such as migration to quantum-resistant cryptography. In decentralised networks, upgrades require consensus among a global and diverse set of participants (e.g. miners, node operators, developers and users), making the implementation of new cryptographic standards slower and more complex than in centralised systems.

This governance challenge amplifies risk in ways that extend beyond quantum threats. For example, 51% attacks, which are where an entity gains majority control of a network's mining power, highlight the vulnerabilities inherent in decentralised **consensus** mechanisms. However, unlike quantum risks, these attacks are not exacerbated by advancements in quantum computing, as they rely on classical computational dominance and network dynamics. Even if quantum-safe algorithms exist, deploying them across millions of wallets and nodes remains a multi-year, resource-intensive effort, dependent on both technical adoption and community consensus.

IV. Moving toward a safer ecosystem

Cryptocurrencies rely on a delicate balance of cryptographic assumptions and decentralised governance. While the blockchain's integrity is largely secure in the near term, the security of individual wallets and the trust in ownership of digital assets is vulnerable to potential advances in quantum computing. Understanding this distinction is essential. The primary risk is not systemic collapse but the exposure of funds and the loss of trust, which could have broader implications for market confidence and financial stability.

Nonetheless, while the development of quantum computers is already well under way, research into countermeasures has already put forward a set of solid viable options to protect current cryptographic standards, to upgrade and/or to integrate them with quantum technology. In the following section, we describe three subsets of possible countermeasures and discuss their viability.

a. Incremental upgrades for quantum resistance

A first short-term solution to achieve quantum resistance focuses on structural and procedural improvements within existing blockchain frameworks, avoiding reliance on post-quantum cryptography (PQC). Instead, it leverages account abstraction, transaction validation methods and address management to reduce exposure to quantum threats.

A core challenge in blockchain security is the use of externally owned accounts (EOAs), traditional wallets controlled by private keys that directly expose public keys on-chain. Account abstraction, as proposed in Ethereum's ERC-4337 standard[23], replaces EOAs with smart contract wallets, where transactions are executed by code rather than private keys. This design introduces a layer of indirection, reducing direct exposure of private keys, and enables integration with zero-knowledge proofs (e.g. ZK-STARKs[24]). ZK-STARKs validate transactions without revealing underlying key information, limiting the utility of harvested public keys for future decryption.

However, this raises the critical question about wallets whose public keys have already been exposed. For these, no cryptographic fix exists; once a public key is exposed, it remains permanently vulnerable to future quantum decryption. The only solution is pre-emptive migration. Users must transfer funds to new, unexposed wallets (e.g. smart contract wallets) before any quantum attack occurs. Exposed EOAs cannot be secured retroactively. They must be abandoned in favour of quantum-resistant alternatives.

Another layer of defence is systematic address rotation, where wallets automatically generate new public/private key pairs for each transaction. This forces adversaries to repeatedly re-target keys, rendering attacks impractical if keys are rotated faster than quantum computers can decrypt them. However, this method assumes quantum computers cannot achieve instant decryption, **a risk that grows with technological progress.**

23 https://docs.erc4337.io/core-standards/erc-4337.html

24 What is ZK-Stark?, StarkWare

Additionally, frequent rotation could disrupt usability or strain network performance if not carefully managed.

In the event of a confirmed quantum attack, a hard fork could revert the blockchain to a pre-compromise state and disable vulnerable EOAs, mandating all transactions to use smart contract wallets with ZK-STARK validation. This is a reactive measure, however, and does not protect funds already spent from compromised wallets.

These conventional methods provide **immediate, practical mitigation**, but are not without limitations. Their effectiveness depends on **user adoption**. As migration from exposed wallets is not automatic, users must proactively transfer funds to new addresses. Governance also plays a critical role as hard forks and protocol upgrades require **community consensus**, which can be slow and contentious. Lastly, some of the techniques presented above would imply significant protocol changes in existing cryptocurrencies, such as Bitcoin, as they lack the infrastructure for account abstraction or smart contract wallets.

While these strategies **buy time**, they are **not foolproof**. If quantum computing advances faster than expected, **unmigrated funds in exposed wallets remain at risk**. These measures should therefore be viewed as **complementary to long-term solutions like post-quantum cryptography**, not as replacements.

b. Adoption of post-quantum standards

The second approach directly confronts the quantum threat to foundational cryptographic standards such as the elliptic curve digital signature algorithm (ECDSA), which underpin not only the Bitcoin ecosystem but also global digital security infrastructure. With the recognition that this risk extends far beyond cryptocurrencies, threatening everything from financial systems to government communications, massive research and standardisation efforts have been mobilised to develop secure alternatives. Post-quantum cryptography (PQC) has emerged as the leading solution, providing algorithms designed to withstand quantum attacks while remaining compatible with classical hardware. By leveraging mathematically intractable problems, such as lattice-based cryptography, hash-based signatures and code-based schemes, PQC offers a future-proof foundation for long-term security in both blockchain and traditional systems.



In 2024, the U.S. National Institute of Standards and Technology (NIST) finalised its first set of post-quantum cryptography (PQC) standards[25], including CRYSTALS-Kyber[26] (key encapsulation) and CRYSTALS-Dilithium[27] (digital signatures), with a second standardisation phase[28] underway to diversify options beyond lattice-based solutions. This transition is not just a technical upgrade but a geopolitical priority, driven by coordinated

25 **Announcing Issuance of Federal Information Processing Standards (FIPS) FIPS 203, Module-Lattice-Based Key-Encapsulation Mechanism Standard, FIPS 204, Module-Lattice-Based Digital Signature Standard, and FIPS 205, Stateless Hash-Based Digital Signature Standard**, NIST, 2024

26 FIPS 203, Module-Lattice-Based Key-Encapsulation Mechanism Standard, NIST, 2024, **https://doi.org/10.6028/NIST.FIPS.203**

27 FIPS 204, Module-Lattice-Based Digital Signature Standard, NIST, 2024, **https://doi.org/10.6028/NIST.FIPS.204**

28 **NIST Announces 14 Candidates to Advance to the Second Round of the Additional Digital Signatures for the Post-Quantum Cryptography Standardization Process**, NIST 2024

efforts across governments and standards bodies. In the European Union, the Coordinated Implementation Roadmap for the Transition to Post-Quantum Cryptography[29] is backed by national cybersecurity agencies, such as:

- **Germany's BSI**[30] (Federal Office for Information Security), which has issued guidelines for PQC migration in critical infrastructure.
- **The Netherlands' AIVD** (General Intelligence and Security Service), together with the CWI and the TNO, have released a PQC migration handbook[31].
- **France's ANSSI** (National Agency for the Security of Information Systems) has released its views on the post-quantum cryptography transition[32].
- **The UK's NCSC** (National Cyber Security Centre), which, despite Brexit, aligns with EU efforts through its quantum-safe cryptography initiatives.

Beyond Europe, Japan's CRYPTREC[33] and Canada's CSE[34] (Communications Security Establishment) are also accelerating PQC adoption, while NATO's cybersecurity frameworks[35] now include quantum-resistant requirements for member states. These global mandates reflect a consensus: PQC is no longer optional; it is a necessity for national security and economic stability.

While Bitcoin and Ethereum have yet to transition, smaller projects such as the Quantum Resistant Ledger (QRL)[36] demonstrate that PQC is already viable. This was the first blockchain to fully implement XMSS (eXtended Merkle Signature Scheme), a hash-based, quantum-resistant signature algorithm, proving that PQC can be deployed at scale.

Despite these advancements, scalability and adoption remain challenging. For example, hash-based signatures (e.g. XMSS, Lamport) offer provable quantum resistance but suffer from large signature sizes (approximately 20KB instead of 72 bytes for the ECDSA), risking blockchain overload. Optimisations like Merkle tree-based key compression are under development to address this, but user migration and governance coordination pose even greater challenges.

The transition to PQC is a global imperative, not just a cryptocurrency issue. For blockchains, it demands wallet upgrades, consensus changes and community adoption, a complex but necessary evolution to avert catastrophic breaches. Before exploring the practical strategies and obstacles in achieving this transition, a third potential approach is described.

29 C(2024) 2393 final, COMMISSION RECOMMENDATION of 11.4.2024 on a Coordinated Implementation Roadmap for the transition to Post-Quantum Cryptography, European Commission

30 Quantum Technologies and Quantum-Safe Cryptography, BSI

31 The PQC Migration Handbook, AIVD, CWI, TNO

32 ANSSI views on the Post-Quantum Cryptography transition, ANSSI, 2023

33 Cryptographic Technology Guidelines, Cryptrec

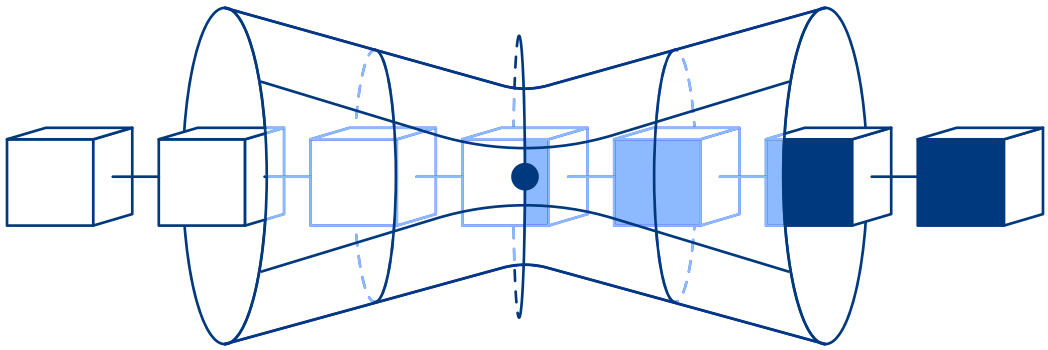
34 Preparing your organisation for the quantum threat to cryptography (ITSAP.00.017)

35 Summary of NATO's Quantum Technologies Strategy

36 https://www.theqrl.org/

c. Using quantum technology to secure the blockchain

Countermeasures such as **quantum cryptography,** which uses principles of quantum mechanics to secure communications, can also leverage quantum technology itself. **Quantum key distribution (QKD)**[37], for example, enables two parties to generate a shared secret key by transmitting photons through fibre-optic cables.



The sender encodes information using polarised filters, while the receiver uses beam splitters to detect photon states. Any eavesdropping attempt disrupts the quantum states, alerting both parties to interference. This makes QKD theoretically **unhackable** under ideal conditions.

However, **practical deployment faces significant challenges**. Current QKD systems are limited to short distances (~500 km) due to photon degradation, requiring expensive repeaters for longer ranges. Proposals like quantum blockchains (e.g. Allende et al., 2021[38]) or quantum cryptocurrencies (e.g. Jogenfors, 2019[39]) remain **experimental and unscalable**. For instance, quantum cryptocurrencies rely on the no-cloning theorem[40] to associate coins with unique quantum states, but their 2GB storage capacity pales in comparison to Bitcoin's 300GB blockchain.

In the near to medium term, classical post-quantum cryptography (e.g. lattice-based algorithms) is far more feasible for securing cryptocurrencies. While quantum-resistant technologies hold **long-term promise**, their current limitations, scalability, cost and infrastructure requirements make them impractical for widespread adoption today. The focus should remain on **transitioning to quantum-safe classical algorithms** and improving governance for large-scale upgrades.

37 Ma, Lijun, M., & Alan, T., Xiao, 'High Speed Quantum Key Distribution Over Optical Fiber Network System', Journal of Research of the National Institute of Standards and Technology, Vol. 114(3), 2009, p. 149, https://doi.org/10.6028/jres.114.010

38 Marcos Allende, Diego López León, Sergio Cerón, Antonio Leal, Adrián Pareja, Marcelo Da Silva, Alejandro Pardo, Duncan Jones, David Worrall, Ben Merriman, Jonathan Gilmore, Nick Kitchener, Salvador E. Venegas-Andraca, 'Quantum-resistance in blockchain networks', arXiv, 2021, https://doi.org/10.48550/arXiv.2106.06640

39 J Jogenfors, 'Quantum bitcoin: an anonymous, distributed, and secure currency secured by the no-cloning theorem of quantum mechanics', IEEE international conference on blockchain and cryptocurrency (ICBC), 2019, https://doi.org/10.1109/BLOC.2019.8751473

40 Wootters, W., Zurek, W., 'A single quantum cannot be cloned', Nature, Vol. 299, 1982, pp. 802–803, https://doi.org/10.1038/299802a0

d. The challenge of the transition to PQC

The most critical challenge in post-quantum migration is not the development of quantum-resistant algorithms, as these already exist, but the execution of a global transition under tight constraints. While symmetric encryption (e.g. AES-256) can be fortified against quantum attacks through key-size adjustments, asymmetric systems, which are the foundation of digital signatures, secure communications and financial transactions, face immediate obsolescence once sufficiently powerful quantum computers arrive.

Unlike traditional IT upgrades, post-quantum cryptography (PQC) adoption demands simultaneous modification of infrastructure, which typically evolves incrementally. Some sectors face acute risks due to 'harvest now, decrypt later' attacks. However, the threat landscape differs by use case: while HNDL poses existential risks to long-term confidential data (e.g. classified communications or medical records), its impact on financial transactions is more nuanced. For payment systems, the real-time interception of transactions, not retrospective decryption, represents the most immediate quantum risk. This distinction slightly eases the time pressure for some use cases but does not eliminate the urgency to begin the transition.

Beyond technical hurdles, the transition faces profound governance and scalability obstacles. Bitcoin's 90% miner consensus requirement for significant protocol changes, historically a source of contention as seen during the 2015-2017 SegWit debates[41], could delay or fragment adoption. Meanwhile, NIST-standardised PQC schemes introduce signature sizes 10 to 120 times larger than the ECDSA, threatening to overload block space, increase fees and slow confirmation times, issues exacerbated by Bitcoin's already strained throughput. Even interim solutions like Taproot[42], which adopted Schnorr signatures in 2021, offer no quantum resistance and remain adopted by less than 1% of unspent transaction outputs (UTXOs) in value, limiting their utility as a bridge.

Assuming that the risk is well understood and that a consensus should be obtained, the transition will still not be immediate. A first and immediate action is the responsibility of wallet owners as wallets must be updated to adopt the changes. Such a process implies not only a software update, but also potentially a hardware update if the hardware (wallet) becomes incompatible with the new consensus. On a different level, the UTXOs will also need to be updated in the blockchain. Each update will require a transaction registered in the blockchain, which will limit the availability of the block for normal transactions. As the block size is already limited, even more so with the potential adoption of larger signatures, this will inevitably have an impact of the transaction time. A recent study from Pont et al.[43] estimates the transition time of all UTXOs to be at least a cumulative continuous downtime of 76 days. It is obviously untenable to fully block any transaction for more than two months. Hence, the transition could be spread over time to maintain a decent level of availability (for example, spread over 300 days by dedicating only 25% of each block to the transition).

Not all UTXOs fall under the same risk category, and this depends on the way they are stored in the ledger[44]. In their new form, the transaction only exposes an ephemeral public key during a short period of time until the transaction is confirmed. A quantum computer powerful enough to derive a private key during this time window could steal the funds in parallel. This scenario, defined by Pont et al. as a 'just-in-time (JIT) quantum attack', underlines the need for a **complete migration** of the cryptocurrency ecosystem **before the advent of practical quantum computers**.

41 https://medium.com/@brenmcma/understanding-segwit-and-the-bitcoin-scaling-debate-c9f7170e9e79

42 https://en.bitcoin.it/wiki/BIP_0341

43 Jamie J. Pont, Joseph J. Kearney, Jack Moyler, Carlos A. Perez-Delgado, Downtime Required for Bitcoin Quantum-Safety, October 2024, https://arxiv.org/pdf/2410.16965v1

44 Will Quantum Computing Break Bitcoin? River Learn

V. Conclusion

The transition to post-quantum cryptography is not just a challenge for cryptocurrencies; it is a **historic opportunity to strengthen their security, resilience and user trust**. Unlike traditional financial systems, where migrations can unfold over years under centralised control, decentralised networks like Bitcoin have the chance to demonstrate that trustless systems can **proactively and collaboratively** evolve to rise to the most advanced technological challenges.

The stakes are clear, but so are the solutions:



Technical action must begin now. Cryptocurrencies don't need to wait for perfect solutions; they can adopt post-quantum schemes in phases, ensuring flexibility for future standards. Bitcoin's Taproot upgrade has already proven that incremental improvements are possible. The next step is integrating quantum-resistant alternatives. Projects like QANplatform and IOTA have already successfully implemented post-quantum signatures, showing that the technology is ready and that what is needed is scalable deployment.



Transparency and education are key to maintaining trust. Developers, exchanges and wallet providers have a unique opportunity to strengthen user relationships by clearly communicating risks, timelines and safeguards. The key is preparing to ensure blockchain networks remain the most secure and innovative in the world. Clear roadmaps, accessible explanations and early miner engagement will prevent panic and ensure smooth adoption.



Collaboration across the ecosystem is essential, but it is also a strength. No single entity can secure Bitcoin or other cryptocurrencies alone. However, by working together, exchanges, wallet providers, miners and developers can create a model of decentralised cooperation that sets a standard for other industries. By prioritising high-risk UTXOs, testing PQC implementations and advocating for regulatory clarity, the goal is to prove that decentralised systems can evolve without sacrificing security or trust.



Cryptocurrencies will not collapse, but their trustless nature demands proactive defence. The quantum threat is not merely a distant possibility – it is a looming deadline that offers an opportunity to prepare. The time to act is now. By upgrading infrastructure, educating users and fostering collaboration, the crypto ecosystem can emerge stronger, demonstrating that decentralised networks can adapt to existential threats while preserving their core principles.

Rather than waiting for quantum computers to arrive, now is the time to build a quantum-resistant future. Cryptocurrencies and blockchain technologies will not disappear because of quantum computing; they will evolve to adapt to the coming risks. Acting today protects against future threats and lays the foundation for a safer, more resilient and more trustworthy crypto ecosystem.

VI. Policy guidelines

01. Ensure policymakers have **access to clear, up-to-date information** on the current state and limitations of quantum computing, to allow them to better estimate/assess when a large-scale quantum computer will be achieved and when it will be able to break current cryptographic systems.

02. **Prioritise discussions on migrating to post-quantum cryptography** (PQC) and other quantum-resistant protocols. Engage a wide range of relevant stakeholders to determine the optimal timing for this migration.

03. **Establish a collaborative working group** led by the European Commission to address quantum computing challenges. Include experts from academia, the private sector, member states and relevant EU agencies (e.g. Europol, ENISA, CERT-EU, EU-AMLA). The group should produce informative assessments tailored to high-level policymakers on a regular basis, to facilitate a coordinated EU response.

04. Policymakers and SMEs should **closely monitor the development of central bank digital currencies** (CBDCs) and their potential impact on blockchain security, and assess their vulnerability to quantum computing threats. In Europe, close cooperation between relevant public and private stakeholders and the European Central Bank (ECB) is crucial for ensuring effective regulatory frameworks and robust security against quantum computing threats.

EUROPOL

QUANTUM COMPUTING AND CRYPTOCURRENCIES – BRIDGING TECHNICAL EXPERTISE AND DECISION-MAKING

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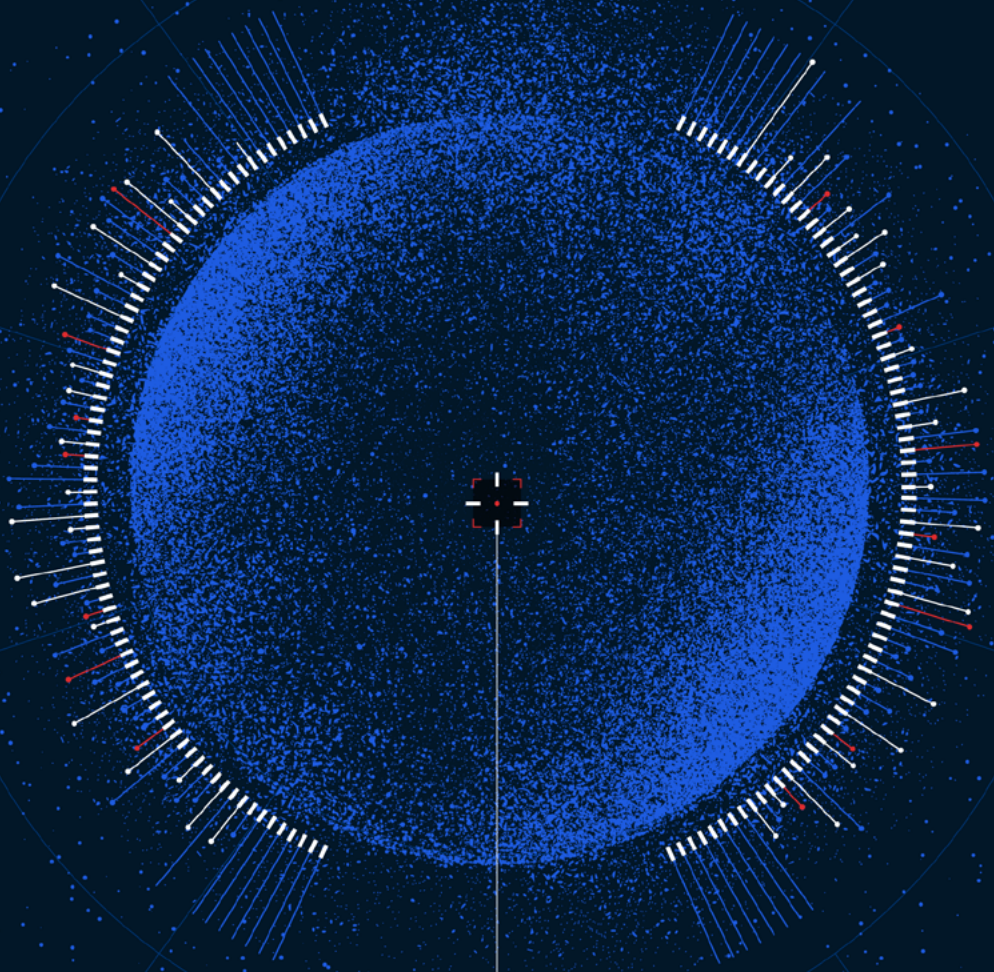
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HARVEST NOW, DECRYPT LATER

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EXECUTIVE SUMMARY

The advent of quantum computing poses a significant threat to current cryptographic systems, particularly those relying on asymmetric encryption, such as RSA and ECC. Quantum algorithms such as Shor's algorithm can efficiently break these systems, making encrypted data vulnerable to decryption in the future. This report explores the 'harvest now, decrypt later' (HNDL) threat, where encrypted data is collected today with the intent to decrypt it later when quantum capabilities become available.

This report, jointly developed by Europol and UC3M within the framework of the Advisory Group Research and Development, which implements Europol's strategy "Delivering security in partnership", assesses the technical feasibility and potential implications of the HNDL threat[1]. It examines possible criminal motivations, identifies categories of data and stakeholders most likely to be affected, and outlines realistic scenarios in which law enforcement authorities may encounter this technique over time.

The report builds on Europol's existing work on quantum technologies, including *The Second Quantum Revolution: The Impact of Quantum Computing and Quantum Technologies on Law Enforcement*[2], the *Quantum-Safe Financial Forum – Call to Action*[3], and the joint report on *Prioritising Post-Quantum Cryptography Migration in Financial Services*[4]. While these publications have primarily focused on raising awareness, supporting preparedness and promoting the transition towards post-quantum cryptography (PQC), this report concentrates specifically on the criminal dimension most closely linked to quantum developments.

In particular, it aims to assess the extent to which the HNDL threat is plausible in the current landscape, and to identify stakeholders that may be more exposed based on the sensitivity and longevity of the data they manage. By doing so, the report seeks to support informed prioritisation of mitigation efforts and to contribute to a better understanding of emerging risks at the intersection of quantum technologies and security.

1 Get to know more about the Advisory Group on Research and Development on https://www.europol.europa.eu/about-europol/european-cybercrime-centre-ec3/ec3-partners

2 Europol, 'The Second Quantum Revolution: the impact of quantum computing and quantum technologies on law enforcement', 11 June 2024, https://www.europol.europa.eu/publications-events/publications/second-quantum-revolution-impact-of-quantum-computing-and-quantum-technologies-law-enforcement.

3 Europol, 'Quantum Safe Financial Forum - A call to action', 7 February 2025, https://www.europol.europa.eu/publications-events/publications/quantum-safe-financial-forum-call-to-action.

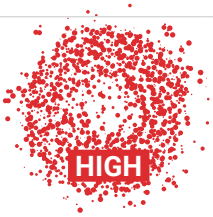
4 Europol, 'Prioritising Post-Quantum Cryptography Migration Activities in Financial Services', 21 January 2026, https://www.europol.europa.eu/media-press/newsroom/news/joint-report-outlines-practical-approach-to-prioritising-post-quantum-cryptography-migration-in-financial-services.

KEY FINDINGS

The HNDL threat is real. It involves collecting and storing encrypted data today, with the expectation of decrypting it in the future using advanced computational methods or quantum computers. This threat does not require the attacker to have a quantum computer today. This threat is particularly relevant for data that retains value over extended periods, such as government communications, health records and intellectual property.

HNDL vulnerability assessment: The technical analysis confirms that standard security protocols like TLS, SSH and OpenPGP, currently in use across the internet, are susceptible to this threat, though risk levels vary based on configurations and key management practices.

STAKEHOLDER IMPACT ASSESSMENT

Stakeholder	Risk	Rationale
Citizens & private individuals		While most personal data have limited long-term value, sensitive information such as financial records (especially for **high-net-worth individuals**), individual medical histories and personally identifiable information (PII) could become attractive targets for identity theft or blackmail in the future.
SMEs & large corporations		Trade secrets, client lists, financial data and sensitive contracts are all high-value assets susceptible to corporate espionage via HNDL. A breach could result in significant financial loss, reputational damage or competitive disadvantage.
Policymakers & government		National security communications (diplomatic cables, intelligence reports), law enforcement case files (ongoing investigations, informant identities), and critical infrastructure data are prime targets for state-sponsored actors utilising HNDL strategies. If compromised, this could jeopardise national security, diplomatic relations or public safety.

RECOMMENDATIONS

LOW-RISK STAKEHOLDERS:

1. **Practise cybersecurity hygiene**:
   Use strong, unique passwords and enable multi-factor authentication. Regularly update devices and software to benefit from vendor security upgrades – including future PQC implementations offered through hybrid cryptographic solutions.

MEDIUM-RISK STAKEHOLDERS:

1. **Prioritise immediate security enhancements**: Upgrade to TLS 1.3 and SSH2, disable legacy protocols and enforce forward secrecy;
2. **Implement data minimisation**: Identify and securely delete unnecessary sensitive data. Limit long-term storage of confidential information;
3. **Invest in long-term resilience** (as resources allow): Explore options for adopting PQC, including hybrid approaches, as it becomes more readily available through vendor solutions and standardised frameworks.

HIGH-RISK STAKEHOLDERS:

1. **Deprecate legacy protocols**: Immediately disable TLS 1.0/1.1 and other outdated cryptographic protocols and enforce forward secrecy;
2. **Accelerate PQC transition**: Prioritise the integration of post-quantum cryptography into critical infrastructure and communication systems, starting with pilot projects, including hybrid cryptographic deployments, to gain experience and minimise disruption;
3. **Develop a national PQC strategy**: Establish a coordinated plan for migrating to PQC across agencies, including funding, standards development and workforce training.

CONCLUSION

The HNDL threat is a forward-looking risk that requires proactive measures today to mitigate future vulnerabilities. By strengthening cryptographic agility and planning for the transition to PQC, organisations can protect sensitive data against the current HNDL threat and any other cryptographic threats that might emerge in the future.

1.

INTRODUCTION: THE IMPACT OF QUANTUM COMPUTING IN CRYPTOGRAPHY

Quantum computing represents a paradigm shift in information processing, leveraging principles from quantum mechanics, the physical domain that governs the behaviour of particles at atomic and subatomic scale.

Classic computers use binary digits or bits which exist in one or two states: 0 or 1. In contrast, quantum computers employ quantum bits, or qubits, which can exist in multiple states simultaneously due to a property known as superposition. This property enables quantum computers to process a vast number of possibilities concurrently, potentially leading to significant speed-ups for certain types of computations.

Another fundamental aspect of quantum computing is entanglement, a phenomenon where the state of one qubit becomes correlated with the state of another, regardless of the distance separating them. This property can be harnessed to perform complex calculations more efficiently than classical computers.

In essence, quantum computers will be able to break encryption by leveraging their superior computational power in integer factorisation, which is crucial for decrypting widely used cryptographic systems.

Cryptography, on the other hand, is the science of keeping information secret and secure. By using cryptography, a message can be scrambled or transformed into an unreadable form, known as encryption, and only people with the right key can unscramble or verify it through decryption. This fundamental science provides data confidentiality, integrity, authenticity and non-repudiation, allowing us to protect our messages, files, and even our identity when navigating the digital world.

In modern digital communication, we primarily use two types of cryptography (see *Figure 1.1*):

- **Symmetric cryptography** uses the same **secret key** to both encrypt and decrypt data, such as AES (advanced encryption standard), to provide confidentiality;
- **Asymmetric cryptography** (also known as public key cryptography) uses two different mathematically linked keys: a **public key**, which is accessible to everyone, and a **private key**, which is held exclusively by the owner. This key pair enables verification of identities online (through digital signatures) and the secure exchange of sensitive or confidential information (encryption with a public key), even between parties who have never met. Notable examples of public key cryptography algorithms include Rivest-Shamir-Adelman (RSA), El Gamal, Elliptic Curve Cryptography (ECC), and Digital Signature Algorithm ((EC)DSA).

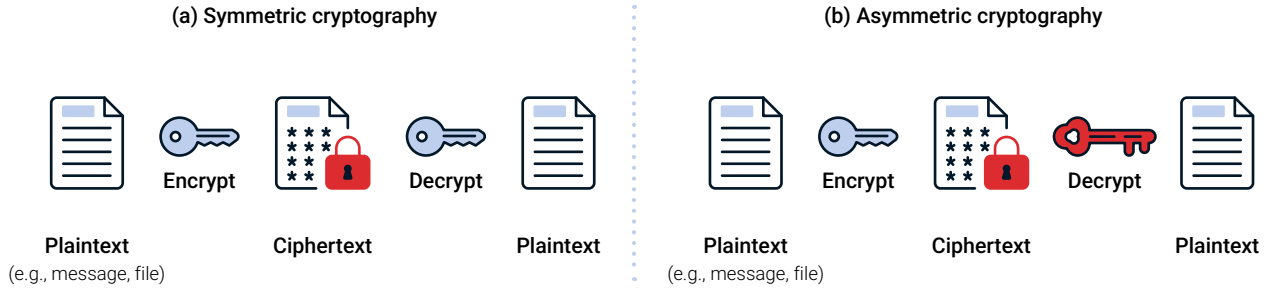


Figure 1.1. Symmetric cryptography (a) vs asymmetric cryptography (b)

In practice, security protocols combine symmetric and asymmetric encryption to secure files and communications. Asymmetric encryption is typically used to exchange a secret key, whereas symmetric encryption uses this secret key to secure (encrypt) the data. The reason for this is efficiency. Depending on the implementation, symmetric encryption can be 1 000 to 10 000 times faster than asymmetric encryption.

1.1. THE QUANTUM THREAT

Generally speaking, cryptography is based on complex mathematical problems that are easy to solve when all required elements are available – typically a secret or private key – but infeasible to crack without it, and requiring thousands to millions of years using classical computers. Asymmetric cryptography security (e.g. RSA) relies on hard problems such as factoring large integers or solving discrete logarithms, while symmetric cryptography (e.g. AES) uses operations such as substitution and permutation to secure data. These foundational problems define the security of cryptographic systems.

However, this foundation is now threatened by the advent of **quantum computing**; quantum algorithms such as **Shor's algorithm** (1994) can solve the specific problems on which asymmetric cryptography relies, with significantly fewer steps than classical computers. In conclusion, current asymmetric cryptography relies on mathematical hurdles that quantum computers overcome far more efficiently, rendering many of today's standards insecure in a post-quantum world.

Unlike current asymmetric cryptography, symmetric cryptography is not fundamentally broken by quantum computers, though it is weakened. **Grover's algorithm** (1996) impacts symmetric algorithms by leveraging quantum computing to perform a brute-force search on an unstructured database with a quadratic speed-up, significantly reducing the time required to crack a key. For instance, an AES-128 key would require approximately 2 to the power of 128 attempts on a classical computer, but it would only require about 2 to the power of 64 attempts on a quantum computer. While this is still a massive number, it effectively halves the bit security strength of the algorithm. A similar principle applies to cryptographic hash functions from the SHA-2 and SHA-3 families, which ensure data integrity. Their collision and pre-image resistance are also impacted by quantum acceleration.

Essentially, quantum computing will be able to **break current asymmetric cryptography** and **weaken current symmetric cryptography**.

1.2. WHEN WILL QUANTUM COMPUTING BE ABLE TO BREAK CRYPTOGRAPHY?

This is often the million-dollar question. However, a definite timeline is less important than it seems. Both cryptography and quantum computing are highly complex fields, and past predictions about their evolution have frequently failed to materialise.

Despite this uncertainty and ongoing scepticism, organisations must still ensure that their information and communication remain secure and interoperable by adopting robust security standards. For this reason, the US National Institute of Standards and Technology (NIST) has published the NIST IR 8547[5] report, which states that classical public key cryptography should be phased out by 2035, with today's most common configurations deprecated by 2030. Similarly, the European Commission has issued a *Recommendation on a Coordinated Implementation Roadmap for the transition to Post-Quantum Cryptography*[6], aimed at guiding the move towards more secure cryptographic solutions.

To address the **potential threat to asymmetric cryptography** posed by quantum computers, global security agencies and cryptography community are working on **post-quantum cryptography** (PQC). PQC relies on new algorithms based on mathematical problems that are believed to be difficult to solve for classical and quantum computers alike. Transitioning to quantum-resistant public key cryptography is essential and requires replacing current system such as RSA and ECC with PQC alternatives. For **symmetric encryption algorithms** and cryptographic **hash functions**, experts recommend precautionary measures such as **larger key sizes and longer hash outputs**, for example migrating from AES-128 to AES-256 or from SHA-256 to SHA-512 where possible.

5 NIST, 'Transition to Post-Quantum Cryptography Standards', 12 November 2024, https://csrc.nist.gov/pubs/ir/8547/ipd.

6 European Commission, 'A Coordinated Implementation Roadmap for the Transition to Post-Quantum Cryptography', 23 June 2025, https://digital-strategy.ec.europa.eu/en/library/coordinated-implementation-roadmap-transition-post-quantum-cryptography.

2.

THE HNDL THREAT

'Harvest now, decrypt later' (HNDL) is a cybersecurity threat that refers to the practice of collecting and storing encrypted data now, with the intention of decrypting it later when cryptographic protections may be weaker or broken. This strategy relies on the assumption that advancements in computing power or new cryptographic vulnerabilities will eventually make it possible to decrypt data that are currently considered secure (encrypted).

Think of a burglar who breaks into an office building and finds a large, heavy safe. The safe is locked, and they do not have the combination yet, so they cannot open it or profit from its contents right away. Instead of walking away, they haul the safe away and store it in a warehouse. Over time, they might learn the combination or acquire better tools to break in.

This raises an important question: will today's information still be valuable tomorrow? Some types of data will undoubtedly retain their value over time, making them attractive targets now as well as in the future. The HNDL threat is a significant concern because it highlights the potential risks of data being compromised long before attackers can decrypt it. Moreover, classical cryptography faces a significant challenge from the development of quantum computers that are expected to break public key cryptography, as demonstrated by Shor's algorithm. In practical terms, this means that information considered secure today will be accessible in the future.

2.1. STEPS INVOLVED IN HNDL



1. DATA COLLECTION (HARVESTING)

Targeting encrypted data: Attackers or adversaries gather encrypted data, often in bulk, from various sources such as intercepted communications, data breaches, network traffic, or online repositories. These data can include personal information such as emails and financial data, but also sensitive corporate information or state secrets.



2. DATA STORAGE

Storing encrypted data: Once the data are harvested, attackers store it for future decryption. The encrypted data are typically stored in a hidden way to avoid detection. This can involve using stolen or throwaway accounts on cloud storage services, other digital services or compromised hosts, peer-to-peer networks or decentralised storage, physical storage devices, etc., to keep the data safe until they can be decrypted.



3. WAITING FOR DECRYPTION

Waiting for cryptographic advances: The attackers wait for new developments in cryptography or computational power that might allow them to decrypt the data. There are two main scenarios criminals might be waiting for:

3.1. ADVANCES IN CRYPTANALYSIS:

Cryptographic researchers might discover weaknesses in encryption algorithms, and new techniques or new vulnerabilities might emerge that make it feasible to break encryption;

3.2. QUANTUM COMPUTING:

One of the main reasons this threat is becoming more relevant is the rise of quantum computing. Quantum computers will break widely used encryption algorithms like RSA and ECC by leveraging quantum algorithms (such as Shor's algorithm) that could factor large numbers or solve discrete logarithm problems much faster than classical computers. Although practical quantum computers capable of breaking encryption may be still far off, adversaries may choose to gather encrypted data in the meantime.



4. DECRYPTION

Decrypting the data: Once attackers have the necessary computational power or have discovered new techniques or vulnerabilities, they can decrypt the harvested data. This may happen after years or even decades.

Once attackers decrypt the stolen data, they can exploit or sell the information for various malicious purposes. This can involve facilitating other crimes, such as financial fraud, identity theft, or corporate/government espionage. Additionally, the stolen data could be used to launch further cyberattacks, blackmail victims, or even influence political outcomes. Organisations, governments and individuals may not even realise their data has been compromised until it is too late.

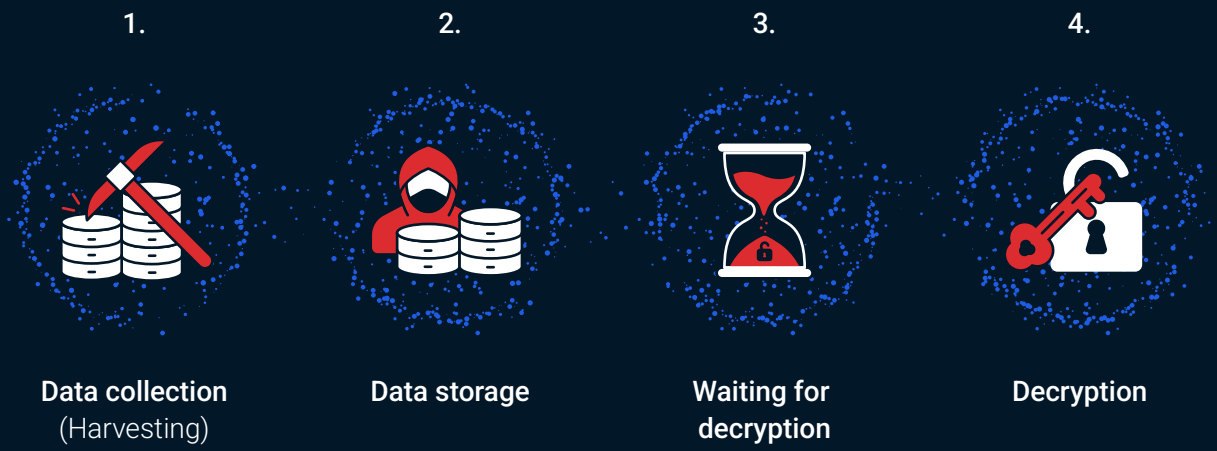


Figure 1.2: Harvest Now Decrypt Later process

2.2. ATTACKER MOTIVATIONS AND STRATEGIC ADVANTAGES OF HNDL

In most cases, cybercriminals aim to obtain an immediate profit from their actions. For example, the initial objective of ransomware activities is to encrypt files to block access to them, forcing companies to pay if they want to recover them and resume operations. Online fraudsters primarily target victims for financial gain, and some data breaches are carried out to sell the stolen information afterwards. But the HNDL threat presents cybercriminals with an alternative, sometimes complementary, strategic approach.

Criminals may resort to the HNDL tactic due to the potential long-term value of encrypted data and the advancement of technology that could eventually render today's cryptography obsolete. While this is a plausible threat, it is difficult to prove the long-term intent of current or potential future criminals. This section will therefore explore the motivations for why a cybercriminal would choose HNDL over (or in addition to) immediate exploitation.



Strategic data accumulation: Some attackers may be focused on accumulating valuable data over time, even if it cannot be immediately accessed or exploited. This is particularly true in espionage or large-scale cybercrime activities like ransomware, where stolen data may not be immediately useful, but could become an asset in the future. Attackers may target data related to intellectual property, personal information (for blackmail purposes) or national security, all of which could become valuable over time.

Think about how collectors operate. An antique or art collector invests in pieces not necessarily for immediate use, but for their potential appreciation in value. They research provenance, condition and market trends to identify pieces that will be worth more in the future. HNDL allows cyber criminals to act like collectors, identifying data with long-term value (like a rare antique) and preserving it for future profit or strategic advantage.



Reducing risk of immediate detection: By passively harvesting encrypted data rather than actively attempting to decrypt it, attackers can avoid triggering security alarms or alerting victims through ransom demands or extortion attempts. At the time of collection, security systems may fail to detect the full scope of the breach, and data might not be easily recognised as valuable, allowing attackers to remain under the radar. Imagine you are in a bustling café. If someone passively eavesdrops on your conversation, you might not even realise they are there, and neither would anyone else. However, if someone actively intercepts your conversation, it is like they are loudly repeating your words to someone else at a nearby table. You would likely notice the interruption, and so would others. Cybercriminals might prefer passive interception because it resembles the quiet listener in the café. They can gather information without alerting their targets or leaving obvious traces of their presence.

HNDL represents a calculated risk, allowing attackers to position themselves for future opportunities. The threat of quantum computing further amplifies the value of harvested data, making this strategic approach a potentially lucrative investment.

2.3. EXAMPLES OF REAL-WORLD CASES THAT COULD USE THE HNDL APPROACH

While there are no widely publicised, specific 'HNDL' cases, there are several real-world incidents where this tactic could be effective. Elements of this approach can be observed more broadly in cyber espionage and ransomware attacks, where attackers harvest and store data with the intention of exploiting it later.

CYBER ESPIONAGE AND DATA THEFT BY NATION STATES

Nation state actors often target intellectual property, sensitive communications, or strategic business or government data. They may not have an immediate use for all the data they steal but can choose to 'store' it for later use, especially as new vulnerabilities or cryptographic breakthroughs emerge.

USE OF RANSOMWARE

What happened: Ransomware groups typically encrypt data on victim systems and demand payment in exchange for a decryption key. In many cases, attackers also exfiltrate sensitive data before encryption and threaten to publish or sell it if the ransom is not paid. Stolen information may be retained for future exploitation, including resale on illicit markets or renewed extortion attempts. In some scenarios, attackers could store encrypted or sensitive data for long periods, potentially exploiting it later if new decryption capabilities emerge or if circumstances increase the likelihood that the victim will pay.

REAL CASE / HNDL ELEMENT

Groups like REvil and Clop have been observed performing triple extortion attacks, which involve:

1. Exfiltrating sensitive data before encrypting the victim's files;
2. Encrypting the victim's files and demanding a ransom for decryption;
3. Threatening to release or sell the exfiltrated data if the ransom is not paid.

The HNDL element comes into play when the attackers store the exfiltrated data, including the encrypted information, for potential future exploitation, rather than immediately monetising it. This approach allows ransomware groups to maximise long-term gain from the attack, combining immediate disruption with strategic future opportunities.

3.

PRACTICAL FEASIBILITY ANALYSIS OF THE HNDL THREAT

In this section, a feasibility analysis of the HNDL threat is conducted across widely used **secure communication protocols**, such as Transport Layer Security (**TLS**) versions 1.2 and 1.3 and Secure Shell (**SSH**), and **OpenPGP-based**[7] **storage of information**. The analysis is based on the paper published, with technical details[8]. The specific conditions that must be met to exploit HNDL are outlined through certain representative use cases.

TLS is primarily used to secure web communications, email, databases or virtual private networks (VPNs). Similarly, SSH is mainly used for remote access and secure file transfer, making it a standard tool for IT system administrators. Finally, PGP-based storage is commonly employed by password managers, backup systems and tools for encrypting files or email archives.

3.1. SECURE COMMUNICATION PROTOCOLS: TLS AND SSH

Although TLS and SSH are used for different purposes, both rely on asymmetric cryptography at the beginning of the communication to 'shake hands' and set security rules between client and server. This process is called the security negotiation phase, and it usually involves two main steps:

1. **Identity verification** (or authentication), using the client's and/or server's private keys (i.e. digital signatures or proof of possession of the private key associated with a public key certificate), without revealing the keys themselves;
2. **Secret key creation**, both sides create a shared secret, the 'session key', which will be used to encrypt the exchanged data.

Once the security negotiation phase finishes, client and server switch to fast symmetric encryption, ensuring applications run smoothly and quickly.

A standardised and widely used method of creating a shared secret key (or session key), over the open internet, is the Diffie–Hellman (DH)[9] method. In DH, the key itself is not sent. Instead, each party generates a private secret key and corresponding public value. They exchange their public values, and each party combines the other party's public value with their own private secret key. When both sides perform this operation, they generate the same shared key without ever having transmitted their confidential information or the shared secret key itself, as shown in Figure 3.1.

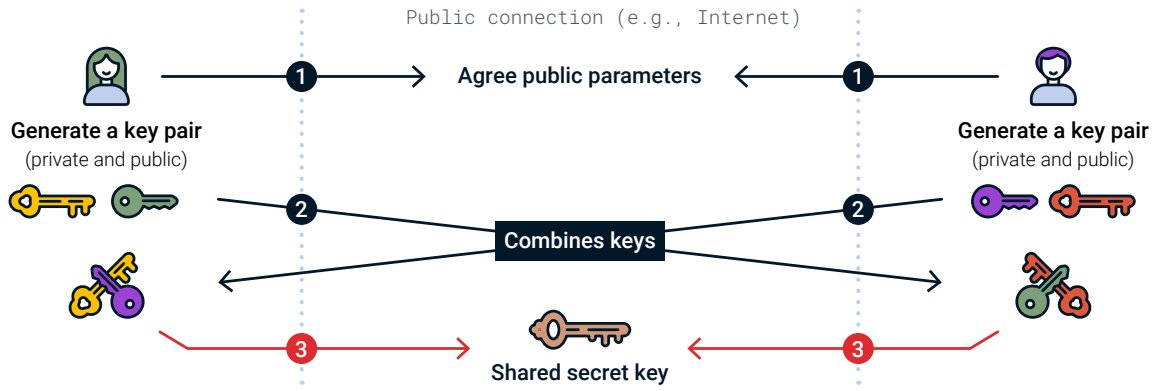


Figure 3.1. Diffie–Hellman Key Agreement

7 OpenPGP website, https://www.openpgp.org/.

8 Blanco-Romero, J., Mendoza, F.A., Rubio, C.G., Campo, C. and Sánchez, D.D. (Cornell University), 'On the Practical Feasibility of Harvest-Now, Decrypt-Later Attacks', 1 March 2026, http://arxiv.org/abs/2603.01091.

9 Rescorla, E., 'Diffie–Hellman Key Agreement Method', *IETF RFC 2631*, June 1999.

In TLS, this occurs during the **handshake protocol**, where client and server negotiate supported cipher suites (cryptographic algorithms), exchange cryptographic material (using DH), and authenticate the server (and optionally the client) using certificates tied to public/private key pairs. The outcome of this handshake is the derivation of shared symmetric keys used to secure the communication channel efficiently.

Figure 3.2 illustrates the full handshake mechanisms of TLS versions 1.2 and 1.3, which are recommended today. In both versions, the negotiation begins with a `ClientHello` message (from the Client), followed by a `ServerHello` message (from the Server); however, TLS 1.3 introduces a significant architectural shift to optimise both performance and security. By including key share extensions directly in the initial `ClientHello` message, TLS 1.3 allows the server to derive session keys immediately, reducing the handshake to a trip, compared to the two trips required by TLS 1.2. Furthermore, TLS 1.3 provides enhanced security by encrypting the handshake metadata from the `EncryptedExtensions` message, which includes the server's identity. Finally, TLS 1.3 strengthens the protocol by removing legacy, insecure ciphers and mandating key exchanges based on modern elliptic curves as a fundamental requirement.

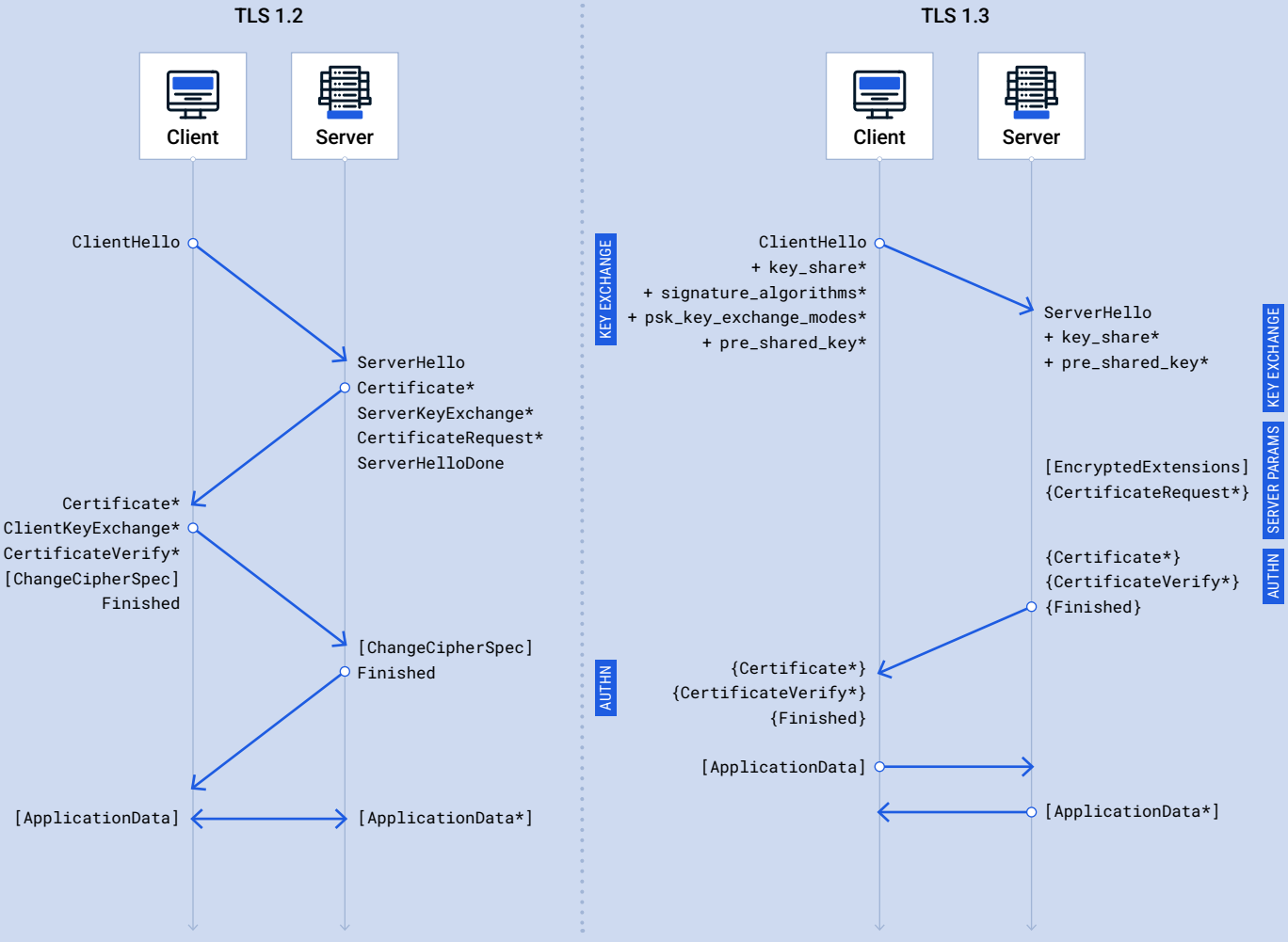


Figure 3.2. Message flow for a full handshake (TLS 1.2 (left)[10] and TLS 1.3 (right)[11])

During this message flow, several intermediary secret keys are derived before generating the final session keys.

10 T. Dierks and E. Rescorla, The Transport Layer Security (TLS) Protocol Version 1.2, RFC 5246, IETF, August 2008, https://www.rfc-editor.org/rfc/rfc524

11 E. Rescorla, The Transport Layer Security (TLS) Protocol Version 1.3, RFC 8446, IETF, August 2018, https://www.rfc-editor.org/rfc/rfc8446

Similarly, as shown in Figure 3.3, SSH begins its connection establishment with a key exchange phase in which client and server agree on algorithms (`KEXINIT` message), perform an exchange based on DH (`KEXDH` and `NEWKEYS` messages), and authenticate the server's identity through its host key. Optionally, user authentication often relies on asymmetric key pairs (`USERAUTH` messages), eliminating password-based weaknesses. Once user authentication is successful, information is encrypted and exchanged (Application Data).

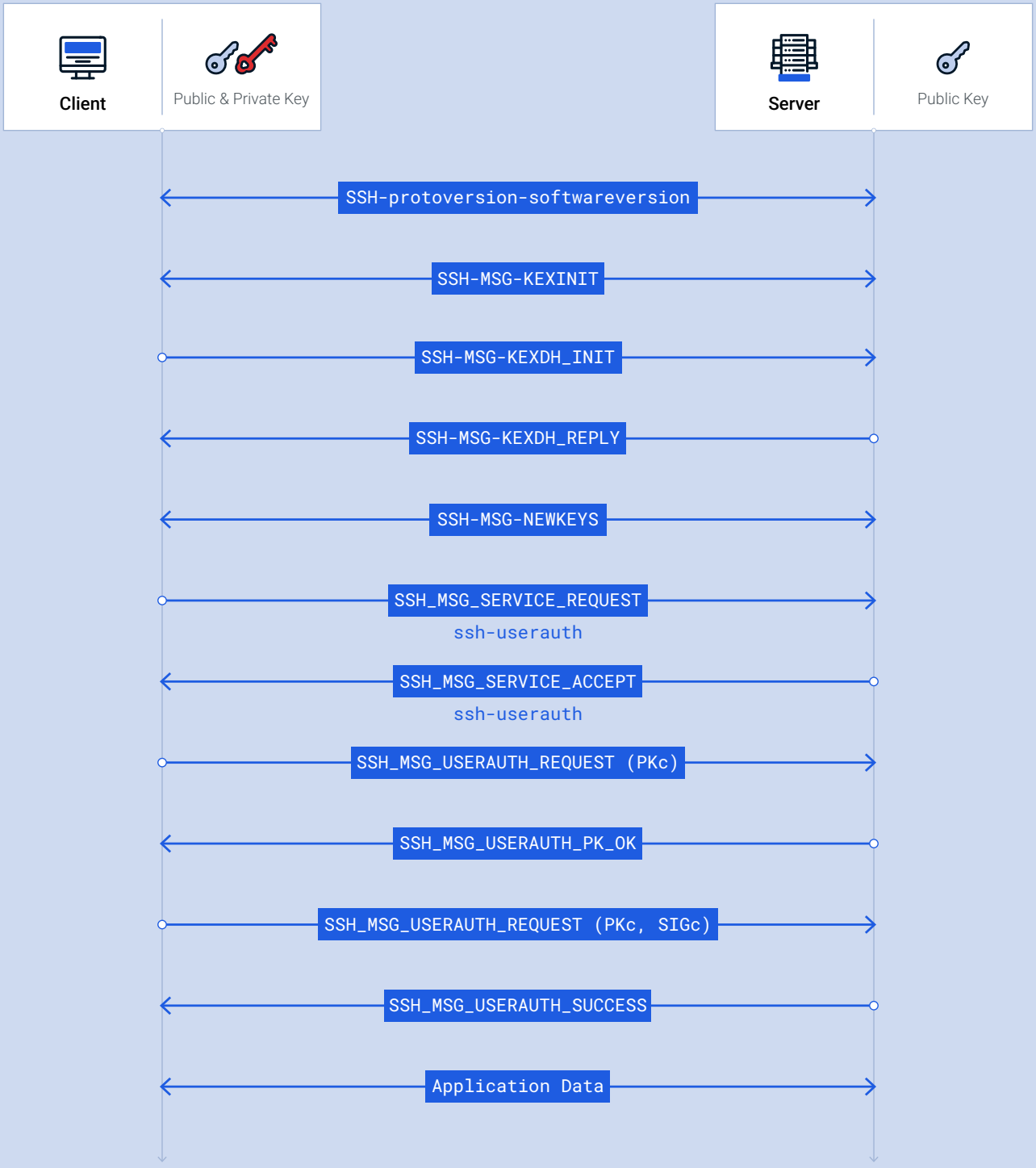


Figure 3.3. SSH Message flow using DH and public key cryptography for user authentication[12]

12 T. Ylonen and C. Lonvick, The Secure Shell (SSH) Transport Layer Protocol, RFC 4253, IETF, January 2006, and The Secure Shell (SSH) Authentication Protocol, RFC 4252, IETF, January 2006, subsequently updated by later RFCs.

3.2. STORAGE WITH OPENPGP

HNDL is not limited to network communications' 'data in transit'; it can also apply to encrypted files' 'data at rest'. An adversary might collect encrypted files, archives or database dumps today, hoping to decrypt them later when keys are compromised. One common example is encrypted email or documents protected with OpenPGP.

OpenPGP is the open standard for PGP (Pretty Good Privacy), which utilises asymmetric cryptography to protect data at rest and in transit, ensuring confidentiality, integrity and authenticity. PGP generates a random symmetric session key (using algorithms such as AES) to encrypt the actual data (e.g. a message or a file). This session key is then encrypted using the public key of the intended recipient and bundled with the ciphertext (see Figure 3.4). Upon receiving the file, the recipient uses their private key to decrypt the session key, which in turn decrypts the data. This hybrid scheme allows secure large files and storage volumes without the significant performance penalties associated with pure public key cryptosystems.

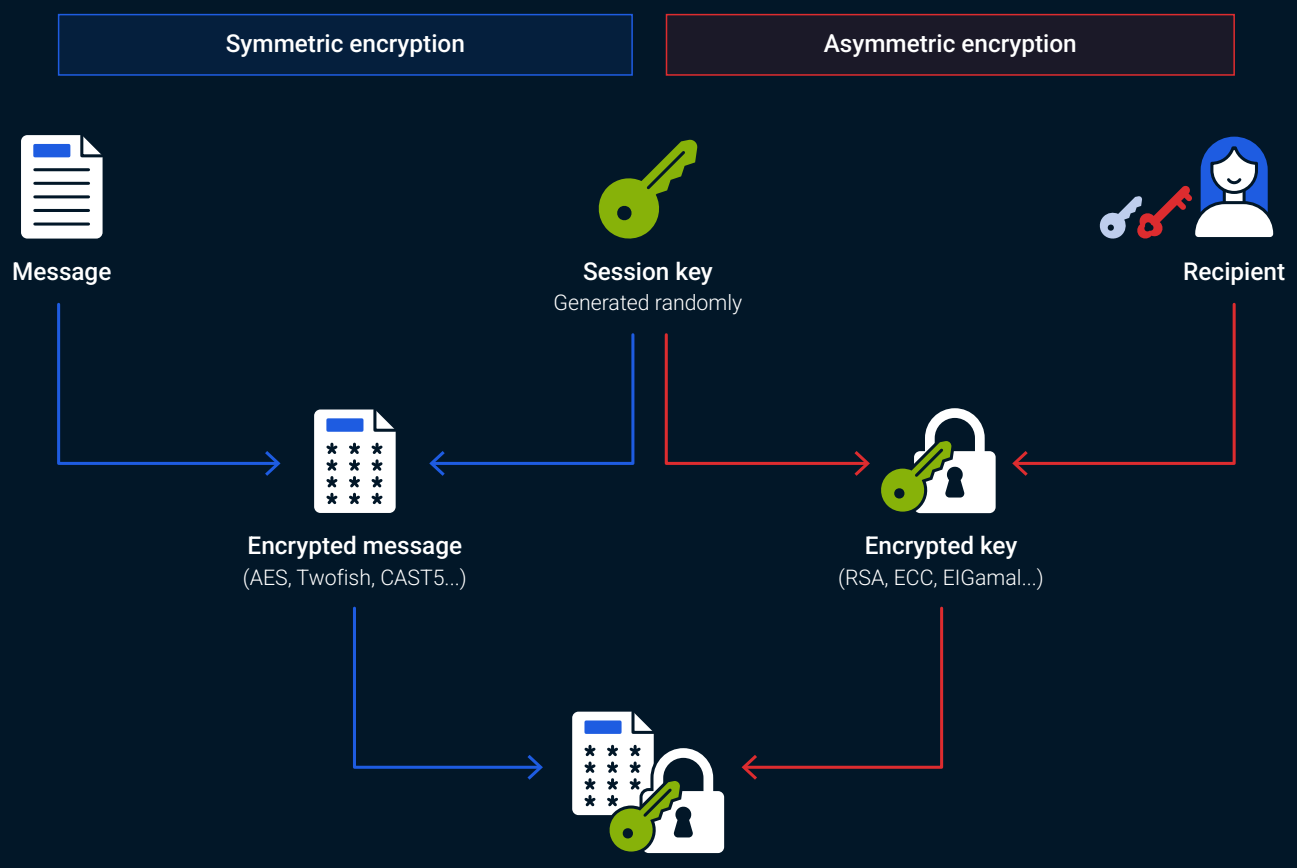


Figure 3.4. PGP Hybrid Encryption

This report focuses on PGP-based file encryption as a representative storage encryption scenario, and it analyses what an attacker would need to decrypt such data later.

3.3. HNDL VULNERABILITY ANALYSIS FOR TLS

To perform a HNDL attack in TLS, a passive adversary records the entire handshake and encrypted traffic of a session, then later uses quantum computing (or other cryptanalytic breakthroughs) to get private keys and break the handshake's cryptography and recover the session keys.

The data required to break the handshake's cryptography will depend on the TLS version and the type of key exchange used. In general, the key exchange methods used in TLS1.2 and TLS1.3 can be grouped into two types:

- **Non-forward secrecy key exchange**:
  In these types of key exchange, the private key of the server is reused, meaning that if we manage to break it, we will be able to decrypt past and future handshakes without breaking any other key.

- **Perfect forward secrecy (PFS) key exchange**:
  In these types of key exchanges, the private keys of the server are not reused, or even used at all during the handshake. A new pair of public/private keys is used for each exchange, usually referred as ephemeral keys. In practice, it means that the attacker will have to break a private key for each handshake they want to recover.

TLS 1.2 supports the two types of key exchange, whereas TLS 1.3 only supports PFS key exchange methods, and it is therefore more secure. Meaning that, in TLS 1.2, if the TLS session used a non-forward secrecy key exchange (RSA or other static modes), then obtaining the server's private key can be enough to decrypt past recorded TLS 1.2 sessions.

If the session used forward secrecy, as in TLS 1.2 with ephemeral (DHE), or based on elliptic curves (ECDHE), then the long-term private key alone is not sufficient, and each recorded session would need to be broken individually.

TLS version	WITH forward secrecy	WITHOUT forward secrecy
TLS1.3	TLS_AES_128_GCM_SHA256 TLS_AES_256_GCM_SHA384 TLS_AES_128_CCM_SHA256 TLS_AES_128_CCM_8_SHA256 *TLS_CHACHA20_POLY1305_SHA256* *(Not listed by NIST, but recommended by IETF RFC8446)*	None
TLS1.2	TLS_ECDHE_ECDSA_WITH_* TLS_ECDHE_RSA_WITH_* TLS_DHE_RSA_WITH_* TLS_DHE_DSS_WITH_* (*) AES 128/256 in GCM (recommended) or CBC (allowed) mode, using SHA256/384 *(IETF RFC9325 recommends only AES in GCM mode to provide AEAD)*	TLS_DH_DSS_WITH_* TLS_DH_RSA_WITH_* TLS_ECDH_ECDSA_WITH_* TLS_ECDH_RSA_WITH_* (*) AES 128/256 in CBC or GCM mode, using SHA256/384
TLS1.1 **TLS1.0** *Deprecated*	N/A	N/A

Table 1 TLS versions and cipher suites recommended by NIST SP 800-52 Rev2

In TLS 1.3, the server's private key, associated with the server's public key certificate, is used only for authentication purposes. Therefore, compromising it later does not reveal prior session plaintexts; prior sessions would only be at risk if the attacker later recovers per-session handshake secrets or session resumption secrets (for example, Pre-Shared Keys (PSKs) or resumption tickets stored at an endpoint).

Forward secrecy protects against future compromise of long-term keys, but under the quantum threat model even ephemeral key exchanges become vulnerable, although an attacker would need to break each individual key exchange, significantly increasing the computational overhead. While this makes a full-scale attack more difficult, it is by no means impossible for a sufficiently powerful quantum computer.

Table 1 summarises cipher suites that provide forward secrecy and those that do not. In TLS 1.3 all ciphers provide PFS, because it is not negotiable. In TLS 1.2, cipher suites starting with TLS_DHE or TLS_ECDHE provides it. A detailed analysis can be found in the paper 'On the Practical Feasibility of Harvest Now, Decrypt Later Attacks'[13]. Other minor TLS versions are deprecated because these are vulnerable and considered insecure.

To summarise, both TLS 1.2 and TLS 1.3 are vulnerable to HNDL attacks, but the severity differs. In TLS 1.2 with RSA key transport (non-forward secrecy), compromising a single server key would allow an adversary to decrypt all past and future sessions protected by that key. When ephemeral Diffie–Hellman (DHE or ECDHE) is used instead, TLS 1.2 gains forward secrecy: the attacker would need to break each recorded session's key exchange individually. TLS 1.3

13 Blanco-Romero, J., Mendoza, F.A., Rubio, C.G., Campo, C. and Sánchez, D.D. (Cornell University), 'On the Practical Feasibility of Harvest-Now, Decrypt-Later Attacks', 1 March 2026, http://arxiv.org/abs/2603.01091.

enforces this stronger model by design, removing RSA key transport and mandating ephemeral exchanges for every full handshake, and therefore compromising a server's long-term key alone cannot decrypt past traffic. However, since the ephemeral public values are already present in the recorded handshake, a sufficiently powerful quantum computer could solve the Diffie–Hellman problem to obtain each session's shared secret and decrypt the corresponding traffic; additionally, resumed sessions, relying on pre-shared keys derived from a broken handshake, would also be exposed if the key is not updated (conversely, if the attacker failed to capture the original full handshake, the resumed session remains protected, since the pre-shared key cannot be reconstructed). In short, TLS 1.2 with RSA is the most exposed configuration, whereas TLS 1.3 limits the damage to per-session attacks, although both remain ultimately vulnerable under the quantum threat model.

In both TLS versions, an HNDL attacker would record the entire connection, including the handshake negotiation, all public key exchange values and the encrypted data, even though none of it can be read at that point. The attacker stores this recording and waits. When the necessary cryptographic capability becomes available (whether through key compromise, theft, or a future quantum computer), the attacker can work backwards from the recorded public values to recover the session secrets and decrypt everything. Figure 3.5 illustrates this process for a TLS 1.3 session: once the shared secret from the key exchange is recovered, all session keys follow automatically.

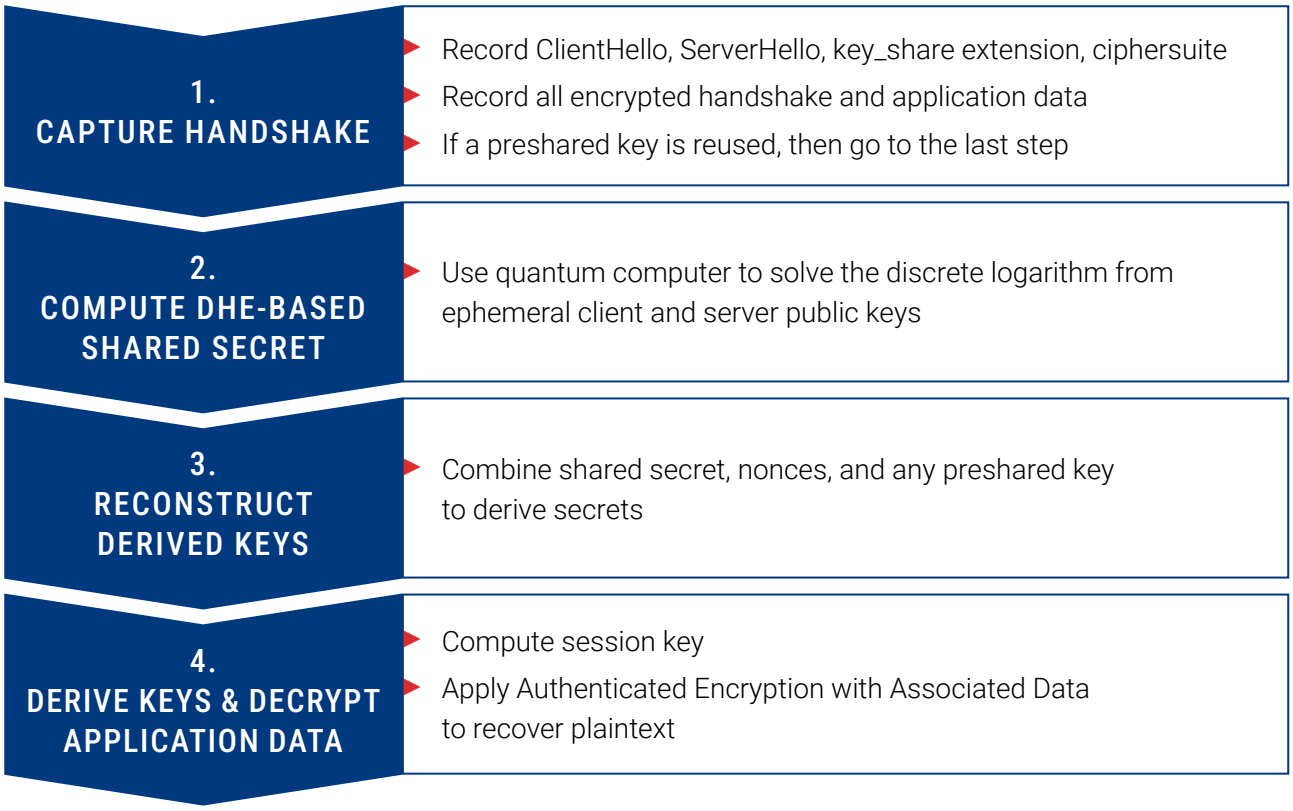


Figure 3.5. Decryption Steps for a Recorded TLS 1.3 Session

Two additional points are worth noting. First, TLS 1.3 offers a 'fast reconnection' mode (called 0-RTT) that reuses secrets from a previous session; if the attacker recorded the original full handshake and later breaks it, the reconnected session is exposed as well. However, if the original handshake was not captured, the reconnected session remains safe. Second, although TLS 1.3 allows endpoints to refresh their encryption keys during a session – something called re-keying – these refreshed keys are derived from the original handshake secret and would therefore also be recoverable once the underlying key exchange is broken.

3.4. HNDL VULNERABILITY ANALYSIS FOR SSH

In the case of SSH communications, under the default and recommended settings, SSH's key exchange is ephemeral and forward-secret. The Diffie–Hellman shared secret key established during the handshake is not stored in the long term; once the session is over and keys are discarded, an attacker cannot recover that secret unless they can solve the DH problem. Importantly, the server host private key (e.g. an RSA, DSA, or key based on elliptic curves (Ed25519)) is used only to authenticate the exchange (by signing certain handshake data) and is not used to encrypt the session key itself. Thus, if an adversary records an SSH session and later steals the server host key, they still cannot derive the session key from the recording. Past SSH sessions remain confidential, illustrating forward secrecy: a compromise of long-term keys does not compromise past session keys. This property protects from HNDL in the classical scenario.

However, as with TLS, the caveat comes with quantum computing. If an attacker can one day break the DH exchange (e.g. using Shor's algorithm to compute the discrete log of the recorded public values), they could then obtain the shared secret key and decrypt it. All widely used SSH key exchange methods (diffie-hellman-group14-sha256, ecdh-sha2-nistp256, curve25519-sha256) rely on the difficulty of the discrete logarithm problem; thus, they would all be vulnerable to a quantum HNDL adversary just as TLS would be. It is worth noting that an older key exchange method for SSH based on RSA (RSA Key Exchange for SSH) involves the server encrypting a secret key with its RSA key, such as TLS 1.2 RSA mode. That method does not provide forward secrecy. However, this RSA-based key exchange has been deprecated and is not commonly used today. All modern SSH connections use the DH family, which provide forward secrecy, including based on elliptic curves; however, as discussed for TLS, a future quantum adversary could still decrypt archived sessions by breaking the recorded Diffie–Hellman exchange.

SSH protocol version	WITH forward secrecy	WITHOUT forward secrecy
SSH-v1 (Obsolete, insecure!)	N/A	N/A
SSH-v2 (Current standard)	RFC 9142 curve25519-sha256 diffie-hell-man-group16-sha512 ecdh-sha2-nistp256 ecdh-sha2-nistp384 ecdh-sha2-nistp521 gss-curve25519-sha256-* gss-group14-sha256-* gss-nistp256-sha256-* gss-nistp384-sha384-* gss-nistp521-sha512-*	

Table 2 SSH version and Key Exchange mechanisms recommended by IETF RFC 9142

Figure 3.6 illustrates the steps of a complete HNDL attack against SSH. A passive adversary captures the SSH handshake, including the SSH_MSG_KEXINIT algorithm negotiation, the SSH_MSG_KEXDH_INIT/REPLY messages containing ephemeral public keys, and all subsequent encrypted traffic. Using a future quantum computer, the attacker solves the elliptic curve discrete logarithm to recover one party's ephemeral private key, then computes the shared secret. With the shared secret and the captured exchange, the attacker applies a hash function to derive all six session keys and decrypt the entire archived session. The single-stage key derivation from the shared secret and hash means all session keys are immediately recoverable once the ECDH shared secret is obtained.

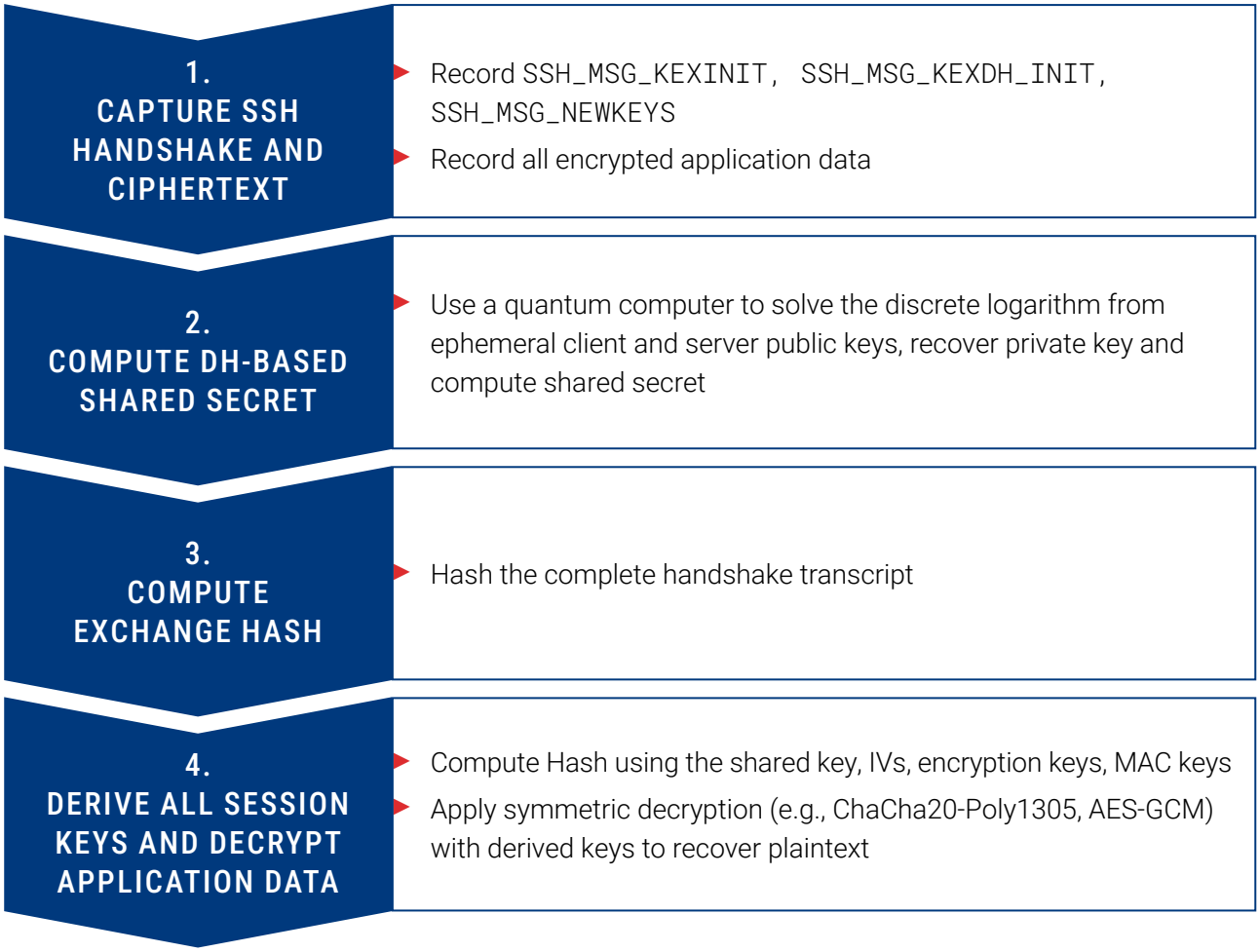


Figure 3.6. Decryption Steps for a recorded SSH session under the HNDL model

Finally, it is worth mentioning that SSH can perform a 're-key' after a connection has started, to refresh its security. It does this by starting a new key exchange (SSH_MSG_KEXINIT) right inside the existing secure connection. This creates a brand-new secret key for that specific part of the session, though the session's 'original ID' stays the same. An HNDL adversary who records the entire session (including re-key exchanges) can break each key exchange independently and derive the corresponding session keys for each epoch.

Unlike a TLS 1.3 key update, which derives new keys deterministically from the previous secret, SSH re-keying performs a full new DH exchange, theoretically requiring the attacker to break each exchange separately. However, this provides no additional quantum resistance, since each re-key has the same underlying ECDH problem.

3.5. HNDL VULNERABILITY ANALYSIS FOR STORAGE

As mentioned before, OpenPGP implements a hybrid encryption scheme: a random symmetric key (also called a session key) is used to encrypt content (with a cipher like AES). Then, that random session key is encrypted with the intended recipient's public key (for example, using RSA or El Gamal encryption) to ensure only the intended recipient (i.e. public key owner) can access it. The result is an encrypted session key, which is stored in the message alongside the ciphertext of the content. Upon receiving it, the recipient uses their private key to decrypt the session key and then uses that session key to decrypt the content.

If an attacker performed an HNDL attack on PGP-encrypted data, two main things would be needed:

1. The **encrypted data itself**, including the encrypted session key packet(s). In practice, this means the attacker must get a copy of the PGP-encrypted file or message. For example, the attacker might intercept an encrypted email in transit or steal a backup that contains PGP-encrypted files.
2. The **corresponding private key of one of the recipients** (along with the ability to use it, i.e. the passphrase or password if it is passphrase-protected) at some point in the future.

Data encrypted with PGP for recipients Alice and Bob will have the following format:

```
-----BEGIN PGP MESSAGE-----
… (some header info)…
<Encrypted Session Key for Alice>
<Encrypted Session Key for Bob>
<Encrypted content data>
-----END PGP MESSAGE-----
```

The encrypted session key for Alice is a blob that only Alice's private key can decrypt (it typically appears as a PKESK packet in the binary encoding), and similarly for Bob. An eavesdropper can see these blobs but cannot decode them without the private key.

Now, suppose the attacker archives this message. Later, an attacker with a quantum computer, having collected the public key of Alice, could derive Alice's private PGP key from her public key. Once the private key is compromised, the attacker can use it to decrypt any session key blob intended for her. That yields the session (symmetric) key that was used to encrypt the content. Finally, the attacker takes the encrypted content and decrypts it with this session key, obtaining the plaintext. This entire process will succeed as long as the cryptographic algorithms remain unbroken and the private key is obtained. There is no additional ephemeral secret key needed, because the 'ephemeral' session key was stored and wrapped by the public key. In essence, the PGP message itself contains all the information needed to decrypt (the content cipher text and the session key encrypted for the private key), and thus possession of the private key at any time will unlock it.

It is also worth noting that PGP private keys are often protected by a passphrase. Consequently, an attacker might obtain an encrypted private key (e.g. by hacking someone's computer or a key server), and then later also guess or crack the passphrase. Once they have the actual private key material, decryption of past messages is straightforward. If the algorithms used for public key encryption (RSA, ElGamal, ECC) are broken through advances (such as quantum computing factoring RSA), then an adversary could derive it from the public key and then do the same even if the private key was not directly stolen. In the context of quantum threats, RSA and ElGamal (which relies on discrete log) would be broken by quantum algorithms, meaning an attacker could potentially compute the recipient's private key from the public key in the future.

4.

MITIGATION STRATEGIES FOR THE HNDL THREAT

Because HNDL targets data that are already being transmitted and stored today, mitigation must be proactive; waiting until quantum computers arrive would be too late for traffic that has already been recorded. The measures available fall into two broad categories: at technical level, organisations can enforce perfect forward secrecy, adopt post-quantum cryptography (PQC), and deploy hybrid key exchanges that combine classical and quantum-resistant algorithms; while at management level they can apply data minimisation, shorter key and data life cycles, and cryptographic agility programmes to prepare for the longer-term transition.

4.1. MITIGATION STRATEGIES AT TECHNICAL LEVEL

4.1.1. SECURE COMMUNICATION PROTOCOLS (DATA IN TRANSIT)

To mitigate HNDL risks, in the short term, the fastest reduction in exposure often comes from using modern protocol versions and secure defaults. Organisations must use the latest and recommended TLS 1.3 and SSH2 versions. Implementations must be updated, and legacy configurations and deprecated protocol versions avoided; that is, organisations must disable TLS 1.0, TLS 1.1, and SSH1 where they are still present. These versions are deprecated due to having known vulnerabilities, being insecure nowadays, and increasing the likelihood of near-term compromise (HNDL).

In addition, some features also require conscious extra caution for TLS:

- **TLS 1.3 0-RTT early data**: this mode does not provide forward secrecy for early data. Must be restricted to low-risk use cases, while avoiding sending sensitive data and using key update.
- **Session resumption and pre-shared keys (PSKs)**: session resumption is widely used for performance, but it needs careful configuration. Where resumption occurs without a fresh key exchange, a future compromise of the resumption secret can expose more than one session's data. (A PSK is a secret key reused to resume or speed up a connection.)

Where TLS 1.2 must be used for compatibility, it should be restricted to forward-secret Diffie–Hellman key exchange modes (DHE or ECDHE) and RSA key transport avoided. This also applies to SSH key exchange modes. Algorithms that use static public/private keys, such as RSA, must be disabled entirely. Although PFS does not defeat a future quantum adversary (who could still solve the ephemeral key exchange), it substantially raises the cost of an HNDL attack: instead of breaking one long-term key to decrypt years of traffic, the attacker must break every recorded session individually.

If possible, the transition to configurations that support PQC should be prioritised (see Section 4.3). Even if full PQC migration is not yet feasible, hybrid key exchanges (see Section 4.4) that combine classical ephemeral DH with quantum-resistant algorithms should be implemented. Nevertheless, for many organisations, migrating to PQC is the long-term goal, but it is not yet consistently deployable across all systems, products and partners.

4.1.2. STORAGE ENCRYPTION (DATA AT REST)

For encrypted data at rest, such as files or archives protected with OpenPGP, the main HNDL risk is that the long-term private key could be compromised or broken by a future quantum computer, exposing every file ever encrypted to that key. In the short term, organisations should ensure that encryption keys are rotated regularly and that older ciphertexts are re-encrypted with fresh keys when practical. Where possible, migrating to PQC-capable encryption tools (for example, GnuPG builds that support post-quantum public key algorithms) will provide longer-term protection. Equally important is limiting the volume of sensitive data stored in encrypted form: data that are no longer needed should be securely deleted rather than archived indefinitely, since every retained ciphertext is a potential future target.

4.1.3. POST-QUANTUM CRYPTOGRAPHY (PQC) ADOPTION

Post-Quantum Cryptography, also known as quantum-resistant cryptography, is a new set of cryptographic algorithms that are secure against both classical and future quantum computers. This can be applied to both data in transit and data at rest. These algorithms use public key schemes either for a key encapsulation mechanism (KEM) or for digital signatures. Unlike key agreement protocols such as Diffie–Hellman (where both parties contribute to the shared secret key), a KEM allows one party to use a public key to generate and wrap a shared secret (encapsulation), and the other party to unwrap it with the corresponding private key (decapsulation). A digital signature defines both the generation and verification of signatures.

KEM algorithms include three main operations: key pair generation, encapsulation and decapsulation. Encapsulation uses a public key to generate a ciphertext and a shared secret key, while decapsulation uses the corresponding private key and the ciphertext to generate the same shared secret key. Signature algorithms also include three operations: key pair generation, signing and verification. Signing uses a private key to sign a message, and verification checks the validity of the signature using the message and the public key.

The algorithms proposed so far are mainly based on five core schemes: lattices, multivariate polynomials, error-correcting codes, hash-based signatures and isogeny of elliptic curves. A summary of the main characteristics of each scheme is provided in the table below. Each algorithm has different versions, with variations in their parameters or encryption and hash functions, which offer different security levels.

Scheme family	Main characteristics
Lattice-based	Flexible construction and good balance between performance, key size, and ciphertext/signature size. Based on computationally hard problems over high-dimensional lattices, such as Learning with Errors (LWE) and Short Integer Solution (SIS) and their variants. Representative examples include CRYSTALS-Kyber for KEM, and CRYSTALS-Dilithium and FALCON for digital signature. CRYSTALS-Kyber and CRYSTALS-Dilithium have already been standardized: FIPS 203: ML-KEM and FIPS 204: ML-DSA, respectively. FALCON-derived FN-DSA standard (FIPS 206) is still under development.

Scheme family	Main characteristics
Multivariate	Specifically investigated for digital signatures. Often trade relatively large public keys for compact signatures and relatively efficient signing and verification. Based on the difficulty of solving systems of multivariate polynomial equations. MAYO, QR-UOV, SNOVA, and UOV (Unbalanced Oil and Vinegar) are currently third-round candidates in NIST's Additional Digital Signature Schemes process. MQOM is also in such round; although NIST classifies it as an MPC-in-the-Head signature, its underlying hard problem is the multivariate quadratic (MQ) problem.
Code-based	Long-standing security foundations, although their key (generally large public key) and ciphertext (generally small ciphertext) vary across constructions. Based on hard decoding problems in error-correcting codes. Representative examples include Classic McElliece, BIKE, and HQC. HQC was selected by NIST for standardization in 2025 following the 4th round of the original PQC process; its final FIPS standard is still pending. SDitH, a third-round additional-signature candidate, uses an MPC-in-the-Head construction based on the syndrome-decoding problem.
Hash-based	Designed exclusively for digital signatures. Compact public and private keys, but comparatively large signatures, involving significant signing and verification costs depending on the construction. Relying primarily on the security properties of cryptographic hash functions. Includes both stateful and stateless variants. SPHINCS+ is an example of a stateless hash-based algorithm, which was standardized by NIST as SLH-DSA in FIPS 205.
Isogeny-based	Despite the collapse of SIDH-based protocols like SIKE (Supersingular Isogeny Key Encapsulation), isogeny-based cryptography remains an active field of interest through candidates like SQISign, which is characterized by very compact public keys and signatures. Based on hard problems involving isogenies between elliptic curves, SQISign is currently a third-round signature candidate, although its compactness comes at the cost of comparatively high computational overhead.

Table 3 PQC algorithms

In addition to the families shown in the table, NIST's third-round additional signature candidates include FAEST, which NIST classifies as a symmetric-based signature scheme. FAEST relies on symmetric-key primitives combined with zero-knowledge/MPC-style techniques and does not naturally fall into the five traditional PQC families considered above.

The implementation and integration of these algorithms should be guided by Mosca's Theorem. Cryptographers use this logic to determine the urgency of the transition: if X (the years your data must remain secret) plus Y (the years needed to upgrade the infrastructure) is greater than Z (the years until a cryptographically relevant quantum computer exists), then the security of the data is already at risk, as shown Figure 4.1.

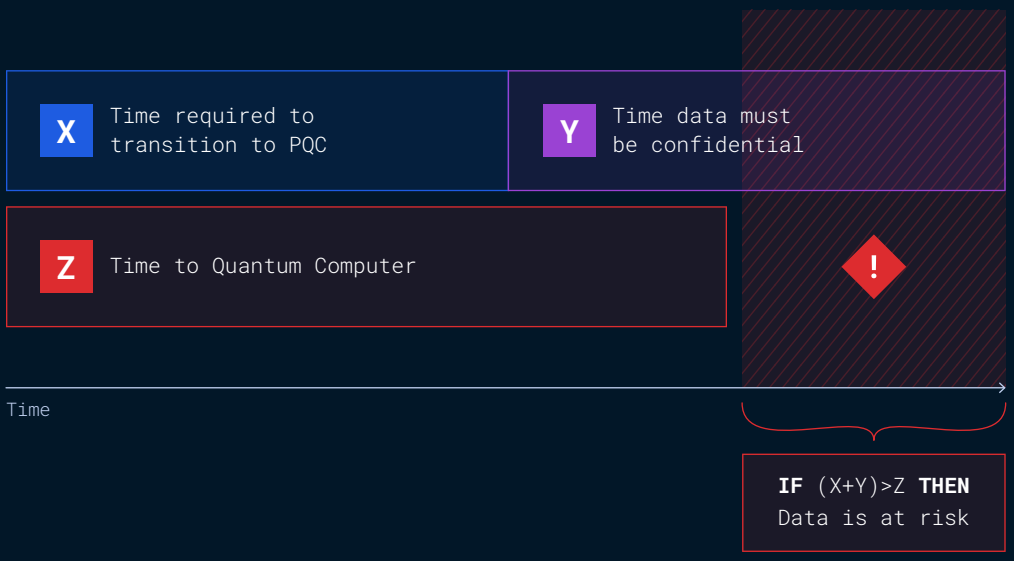


Figure 4.1. Mosca's Theorem

4.1.3.1. PRACTICAL APPLICATION OF PQ ALGORITHMS IN SECURE COMMUNICATION PROTOCOLS

PQC algorithms are available through open-source implementations such as liboqs[14], PQClean[15] and Bouncy Castle[16], among others. These are increasingly integrated into industry-standard secure protocol libraries like OpenSSL[17], wolfSSL[18] and OpenSSH[19], facilitating their adoption within TLS and SSH and other secure communication protocols.

To maintain backward compatibility, most implementations utilise hybrid schemes, combining post-quantum algorithms with classical elliptic curves, by default. Furthermore, other secure communication protocols, for instance VPN protocols such as WireGuard and IPsec, are being updated to support PQC, enabling organisations to establish 'quantum-safe tunnels' or secure connections that protect internal company secrets against both current and future cryptographic threats. Likewise, DTLS is also supporting PQC to be used for IoT protocols (mainly) and other protocols over UDP (User Datagram Protocol).

However, the efficacy of these protocols relies heavily on a globally recognised PQC-based public key infrastructure (PKI) to distribute the digital certificates used to authenticate servers. Consequently, organisations should prioritise inventorying their existing certificates and testing hybrid X.509 certificates in isolated environments; this ensures that legacy network hardware can manage the significantly larger data sizes typical of post-quantum signatures. Parallel to these infrastructure updates, emerging research is exploring alternative architectures that aim to eliminate dependency on traditional digital certificates.

4.1.3.2. PRACTICAL APPLICATION OF PQ ALGORITHMS IN SECURE STORAGE PROTOCOLS

Securing data at rest presents a unique challenge: the encryption must remain robust for the entire lifecycle of the data.

The IETF (Internet Engineering Task Force) is currently finalising the crypto-refresh for OpenPGP to include support for ML-KEM (Kyber) and Composite Signatures. Thus, users can encrypt files using a hybrid scheme where the file is locked with both a traditional RSA/ECC key and a PQ ML-KEM key. This ensures that even if an attacker recovers the RSA private key, for instance in 2035, the ML-KEM layer still protects the file content.

In an equivalent manner, cloud providers that offer 'Zero-Knowledge' encryption (where only the user has the key) are transitioning to PQC to protect user privacy against future state-level decryption. Providers are beginning to use ML-KEM to wrap the symmetric keys (AES-256) used to encrypt large data volumes and ML-DSA to sign file metadata.

14 Github's liboqs page: https://github.com/open-quantum-safe/liboqs.

15 Github's PGClean page https://github.com/PQClean/PQClean.

16 Bouncy Castle's website: https://www.bouncycastle.org/.

17 OpenSSL's website: https://www.openssl.org/.

18 WolfSSL's website: https://www.wolfssl.com/.

19 OpenSSH's website: https://www.openssh.org/.

4.2. MITIGATION STRATEGIES AT MANAGERIAL LEVEL

4.2.1. CRYPTOGRAPHIC AGILITY AND LONG-TERM MIGRATION TO PQC

The measures described above address the HNDL threat at protocol level. However, cryptography is not a one-time deployment: algorithms age, new vulnerabilities emerge, and standards evolve. The overarching lesson of the quantum threat is also that organisations need **cryptographic agility**: the ability to update the algorithms and parameters used across their systems with predictable effort and minimal disruption.

In practice, building cryptographic agility involves several complementary steps:

- First, organisations should inventory the cryptography they currently use. This includes TLS termination points, VPN gateways, SSH servers, backup and archival encryption, email encryption, databases, and any internal services that rely on certificates or key material. Without a clear map of where cryptography is deployed, planning an upgrade is guesswork.
- Second, they should identify which data carry long-term confidentiality requirements. Not all data need to remain secret for decades, but categories such as personal health records, classified government communications, intellectual property and law enforcement case files may well exceed a ten-year sensitivity window, placing them squarely in the HNDL risk zone.
- Third, organisations can reduce the volume of data available to harvest by applying data minimisation principles: enforcing retention limits, securely deleting data that are no longer needed, and segmenting high-value datasets so that a single point of interception does not expose everything.
- Fourth, key management practices should be toughened. Long-term private keys should be stored in hardware security modules (HSMs) where possible, access should be monitored, and incident response plans should cover key compromise scenarios.
- Finally, the transition to PQC should be treated as a programme rather than a single project. Organisations should track vendor and standards body roadmaps (including NIST's post-quantum standards and the European Commission's PQC migration recommendations), test interoperability in non-production environments, adopt hybrid schemes where they are available today, and plan phased rollouts that prioritise the highest-risk systems first.

None of these steps requires waiting for a quantum computer to exist. The goal is to be ready before one arrives, so that the window during which harvested data remain vulnerable is as narrow as possible.

4.3. HYBRID KEY EXCHANGE

Because PQC deployment takes time and must work across diverse systems, many organisations treat **hybrid key exchanges** as a prudent interim approach. This reflects the fact that PQC algorithms are relatively new compared to long-established schemes such as RSA or ECC. In simple terms, a hybrid approach runs two key exchanges in parallel. One is classical (for example, ECDH), and one is post-quantum (for example, ML-KEM). The final session key is derived from both. The intention is that the resulting protection remains strong even if one of the two approaches later proves weaker.

Using hybrid schemes, an attacker would typically need to defeat both the classical and the post-quantum component to recover the session key. The aim is to reduce exposure to HNDL threats without discarding the maturity and operational familiarity of existing classical approaches. However, hybrid deployments add complexity and can introduce interoperability and operational challenges, so they are best treated as a transitional step rather than an end state.

Hybrid key exchange is also moving from research into deployment. In TLS, the IETF has been developing a standard construction for hybrid key exchange in TLS 1.3[20]. IPsec and its key exchange mechanism, IKEv2, can be extended to perform multiple key exchanges and combine secret keys, providing a standards track route to hybrid post-quantum key establishment[21].

Some widely used systems have also started adopting hybrid designs in practice, including OpenSSL and OpenSSH (starting with version 9.0) shipping a hybrid post-quantum and classical key exchange as default[22].

Finally, it is worth noting that another hybrid key exchange, combining a PQC-based key with one generated by quantum nodes, is currently being deployed and tested in highly sensitive environments and research projects. Furthermore, quantum random number generators (QRNGs) are being integrated into key generation processes to begin incorporating quantum computing devices into current scenarios.

20 IETF, 'Hybrid key exchange in TLS 1.3', 20 May 2026, https://datatracker.ietf.org/doc/draft-ietf-tls-hybrid-design/.

21 IETF, 'Multiple Key Exchanges in the Internet Key Exchange Protocol Version 2 (IKEv2)', on the RFC 9370 website, May 2023, https://www.rfc-editor.org/rfc/rfc9370.

22 OpenSSH, 'OpenSSH 9.0 Release Notes', April 2022, https://www.openssh.org/txt/release-9.0.

5.

CONCLUSIONS

This report demonstrates that the HNDL phenomenon presents a credible long-term threat to widely used communication protocols such as TLS and SSH. Under specific conditions, encrypted data intercepted today could be decrypted in the future once sufficiently powerful quantum computing capabilities become available.

However, executing an HNDL attack requires significant resources: large-scale data harvesting and long-term storage, eventual access to a cryptographically relevant quantum computer, and the capability to process and break substantial volumes of key material. Due to these high costs and technical barriers, HNDL is currently mostly viable against high-value targets where long-term confidentiality is critical, such as governmental communications or confidential business information.

Although there is currently no clear evidence that HNDL is being systematically exploited at scale, past cyber incidents involving the exfiltration of encrypted data from high-value organisations suggest that the prerequisites for future decryption may already be in place. Additionally, the risk cannot be indefinitely confined to high-value targets. With quantum capabilities maturing and costs decreasing, the threat may expand to broader sectors.

While HNDL poses a genuine strategic risk, it is not as immediately pervasive as threats such as ransomware. It primarily concerns sensitive data that must remain confidential for extended periods. Notably, the continued use of deprecated protocols such as TLS 1.0 and TLS 1.1 creates an immediate '**Harvest Now, Decrypt Now**' (HNDN) risk, as these versions can be compromised using classical computing techniques without the need for quantum advances.

Overall, organisations should view HNDL not as a speculative concern but as a forward-looking risk that requires proactive cryptographic resilience and migration planning today, to safeguard long-term confidentiality.

MITIGATION AND STRATEGIC RESPONSE

The long-term defence against HNDL is **migration to Post-Quantum Cryptography (PQC)**, designed to resist both quantum and classical attacks. However, mitigation should begin now by strengthening cryptographic agility. Organisations should first inventory where cryptography is deployed – across TLS, VPNs, SSH, certificates, backups, databases and internal services – to understand upgrade dependencies. They should then identify data requiring long-term confidentiality, such as health records, classified communications, trade secrets or law enforcement files, and which fall within the HNDL risk window.

Data minimisation can reduce exposure by limiting retention, securely deleting unnecessary data, and segmenting high-value datasets. At the same time, key management should be toughened through secure storage (e.g. HSMs), strict access controls, monitoring and prepared key rotation procedures.

Finally, the PQC transition should be managed as a phased programme: tracking standards developments, testing hybrid schemes, and prioritising high-risk systems. These measures can and should be implemented now, to minimise the future impact of harvested encrypted data.

While this report provides general guidance on mitigating the HNDL threat, each organisation must conduct its own Quantum Threat Risk Assessment to identify specific vulnerabilities and prioritise mitigation efforts based on its unique data assets, systems and risk tolerance. Developing a phased implementation roadmap aligned with these findings is crucial for effective long-term security.

The quantum threat may be uncertain in its timing, but the imperative is clear: **action must begin now to adopt post-quantum cryptography and secure the future of digital communications**.

GLOSSARY

ASYMMETRIC ENCRYPTION or public key cryptography uses two related keys: a public key and a private key. The public key can be shared. The private key must remain secret. It is widely used for key exchange, digital signatures and identity systems.

AUTHENTICATION is the process of checking or verifying identity. For example, a website may authenticate itself with a certificate, and a user may authenticate with a password, a token or a key.

AUTHENTICITY means you can trust that a person, device, server or message is really what it claims to be. In simple terms, authenticity answers the question: 'Is this really from the right source?'. It helps prevent impersonation, fake websites and forged messages.

CERTIFICATE is a digital document that links a public key to an identity, such as a website name. It is usually issued or signed by a trusted authority. Certificates help browsers and other software decide whether they are talking to the real server.

CONFIDENTIALITY means that information is only visible to the people or systems that are allowed to see it. A common way to protect confidentiality is encryption, which scrambles data so outsiders cannot read it.

DECRYPTION is the reverse of encryption. It uses the correct key to turn ciphertext back into normal readable information.

DIFFIE–HELLMAN (DH) is a way for two parties to agree on a shared secret key over an insecure network without sending the secret key itself through a public channel. That shared secret can then help create a session key. It is a foundational idea behind many modern secure protocols.

DIGITAL SIGNATURE is a cryptographic way to sign a message or document using a private key so that others can verify it with the matching public key. Digital signatures support integrity, authenticity, and often non-repudiation.

EPHEMERAL KEY is a temporary cryptographic key that is created for a short period of time, usually for a single session or connection. It is used during a secure exchange (such as setting up an encrypted internet connection) and then discarded once it is no longer needed.

ENCRYPTION is the process of turning readable information into unreadable data called ciphertext. Only someone with the right key can turn it back into readable form.

FORWARD SECRECY means that if a long-term key is exposed in the future, past recorded sessions should remain protected. This is often

achieved with temporary key exchange methods such as ephemeral Diffie–Hellman.

HNDL describes an attacker strategy in which encrypted traffic or files are collected today and stored for the future. The attacker hopes to decrypt that material later when better tools or more powerful computers become available. This matters particularly when information stays sensitive for many years, such as in the case of government, health, legal or long-term research data.

HASH FUNCTION turns data of any size into a short fixed-size value. Even a tiny change in the original data should produce a quite different hash. Hashes are widely used to check integrity and support signatures and authentication mechanisms.

HOST KEY is the long-term key that an SSH server uses to identify itself to clients. If the host key changes unexpectedly, it may signal a server rebuild, a key rotation or an attack.

HYBRID KEY EXCHANGE combines a traditional method and a post-quantum method in the same connection. The goal is to get the benefits of both during the transition period.

INTEGRITY means that information has not been changed in an unauthorised way. The message you receive should be the same as the message that was sent.

KEM (KEY ENCAPSULATION MECHANISM) is a cryptographic method used to establish or transport a shared secret key securely. Many post-quantum designs use KEMs instead of classic Diffie–Hellman style key exchange.

NON-REPUDIATION means that someone cannot credibly deny an action they performed, such as signing a document or sending a signed message. It usually depends on digital signatures and reliable records.

OPENPGP is a standard and an ecosystem used to encrypt and sign emails, files and messages. It often creates a new session key for each message and then protects that key with public key cryptography. It is widely known from tools such as GnuPG.

PKI is the overall system of certificates, trusted authorities, rules and processes used to manage public keys and digital identity. It is the standardised trust framework behind much of the secure web and many enterprise security systems.

PQC refers to cryptographic algorithms designed to resist both today's normal computers and future quantum attacks that could break some current public key systems.

PRIVATE KEY is the secret partner to a public key. It may be used to decrypt, to sign, or to prove identity, depending on the protocol. If a private key is stolen, the identity or protection linked to that key may be compromised.

PUBLIC KEY is the key that can be shared openly. Depending on the system, it can be used to encrypt data, verify a signature or help establish a secure connection. The key point is not secrecy but trust: people must believe the public key really belongs to the right party.

SESSION KEY is a temporary symmetric key used for one connection, one conversation, or one message. It is usually created at the start of a secure session and then discarded later. Temporary keys reduce the amount of data exposed if a single key is ever compromised.

SSH (SECURE SHELL) is a protocol used for secure remote login, server administration, secure file transfer and encrypted tunnels. It protects command-line access and other remote management tasks.

SYMMETRIC ENCRYPTION uses the same secret key to encrypt and decrypt data. It is fast and efficient, so it is commonly used to protect the actual content of messages and connections. This is why many protocols first agree on a temporary symmetric key and then use it for the rest of the session.

TLS is the protocol that protects many online connections, including HTTPS websites. It helps verify the server, establish session keys and encrypt traffic. When you see a secure website connection, TLS is usually working in the background.

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